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(71) Applicant: **QUALCOMM Incorporated**, San Diego, CA (US)

(72) Inventors: **Andrew WOLVERTON**, Little Elm, TX (US); **Giacomo GHIDINI**, Seattle, WA (US); **David SHAW**, Nazareth, PA (US)

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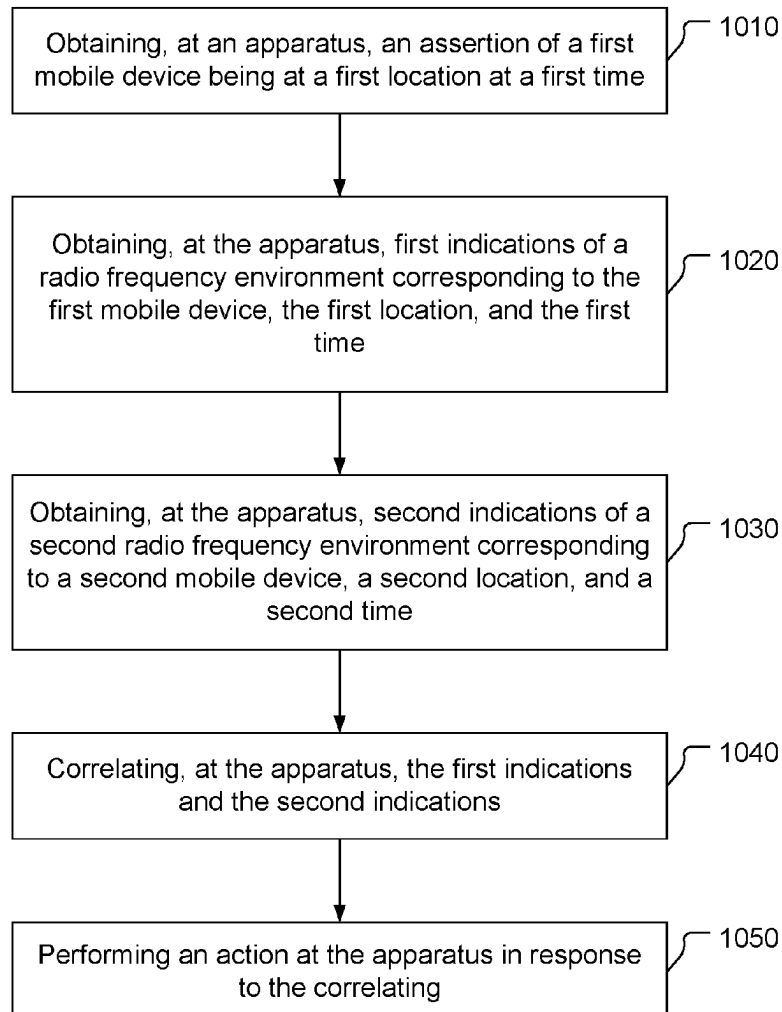
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(57)

ABSTRACT

An example mobile device location assertion verification method includes: obtaining, at an apparatus, an assertion of a first mobile device being at a first location at a first time; obtaining, at the apparatus, first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time; obtaining, at the apparatus, second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time; correlating, at the apparatus, the first indications and the second indications; and performing an action at the apparatus in response to the correlating.

1000



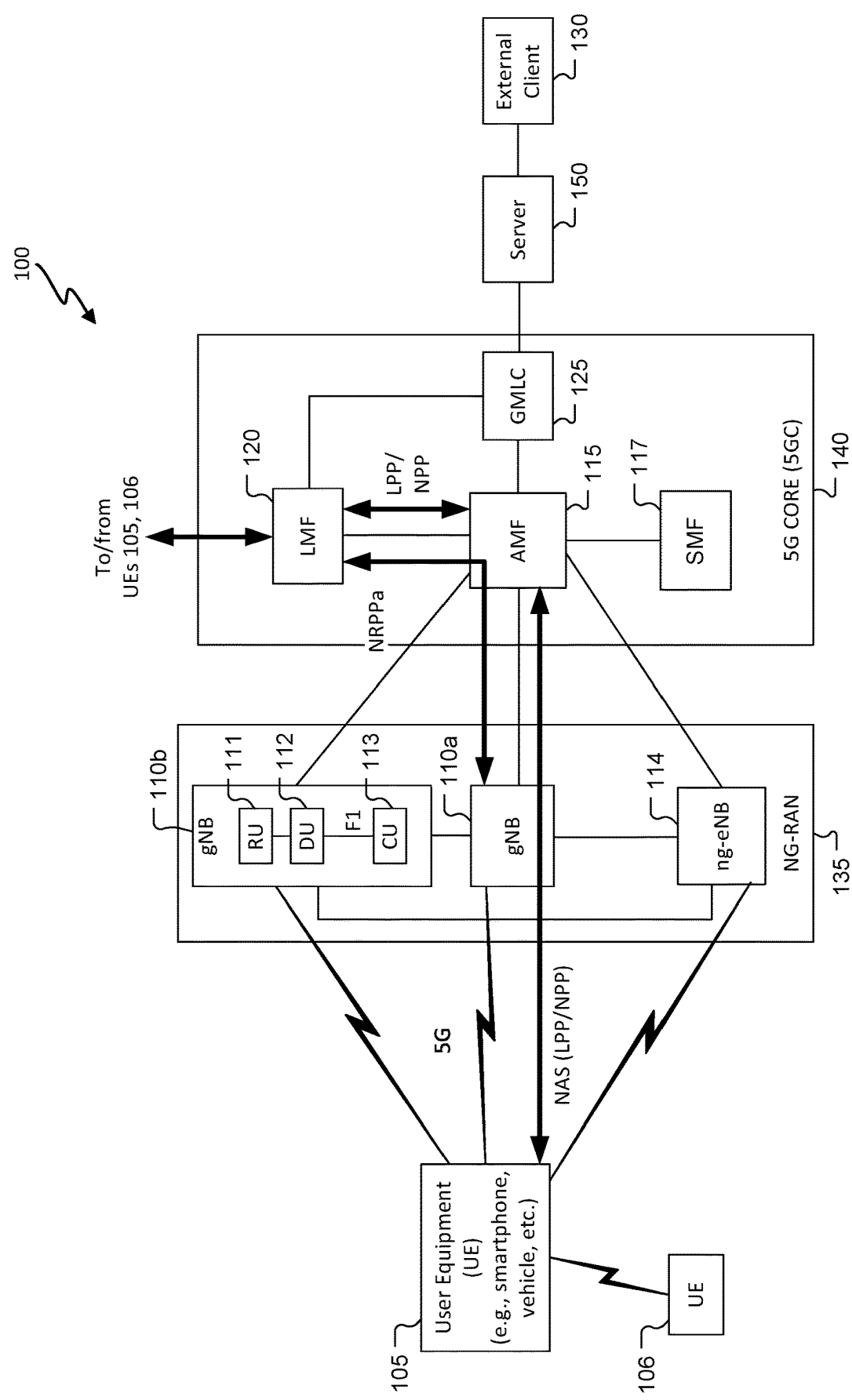


FIG. 1

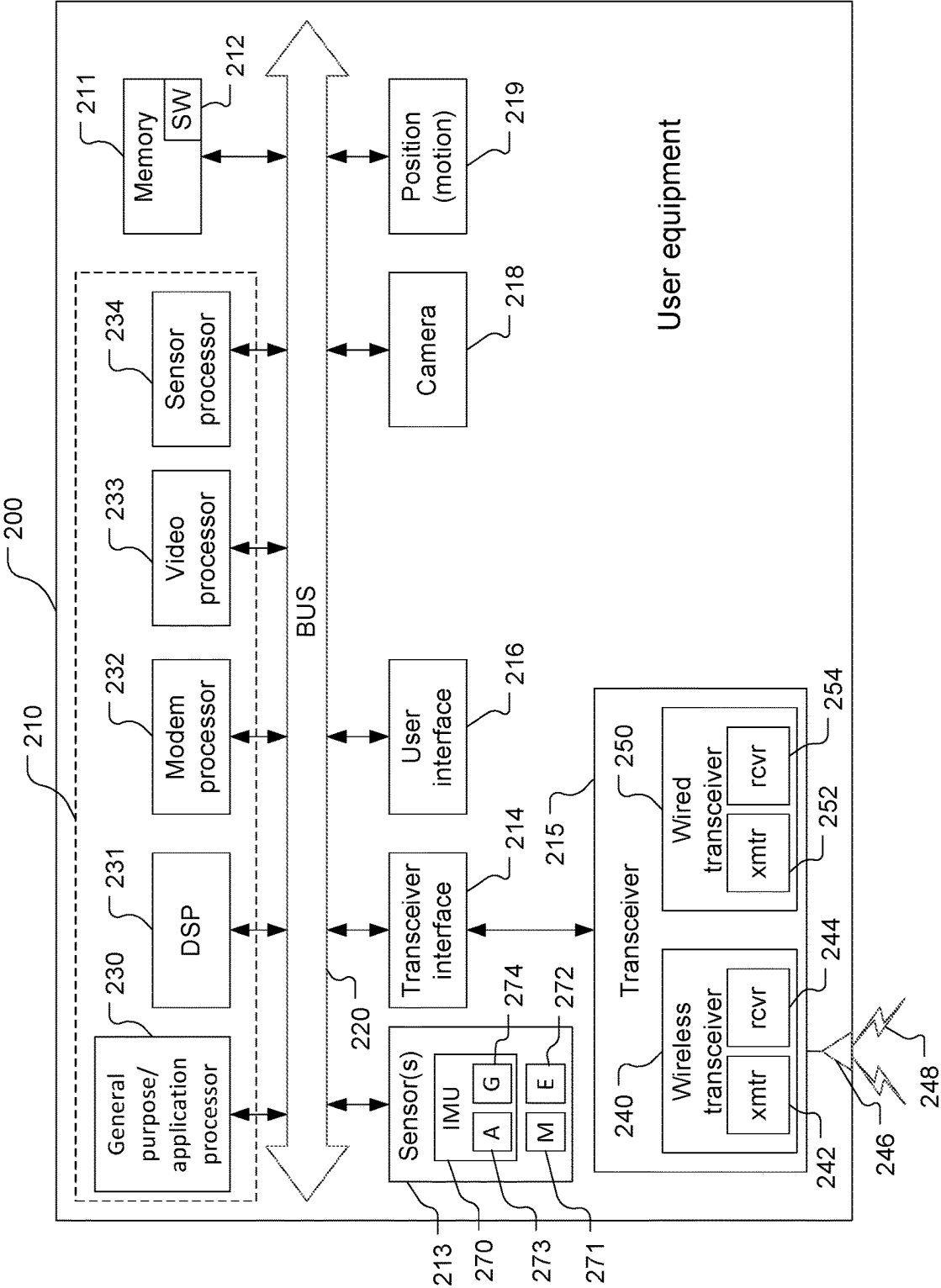


FIG. 2

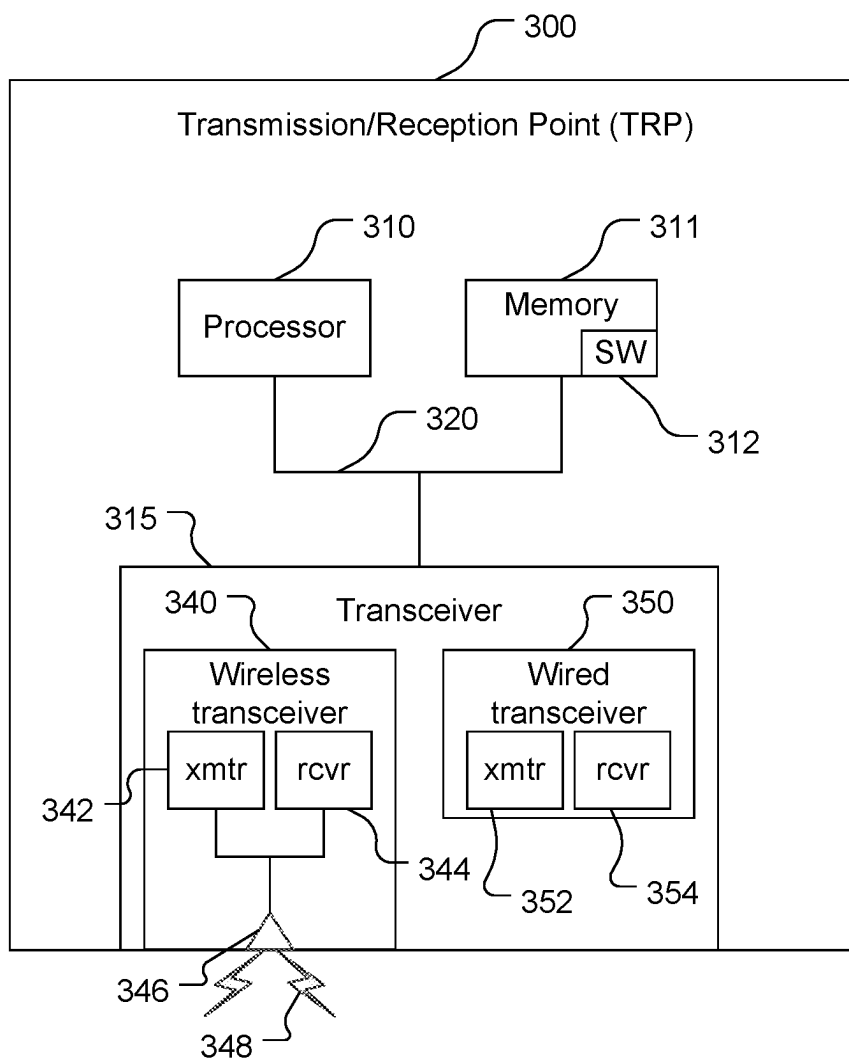


FIG. 3

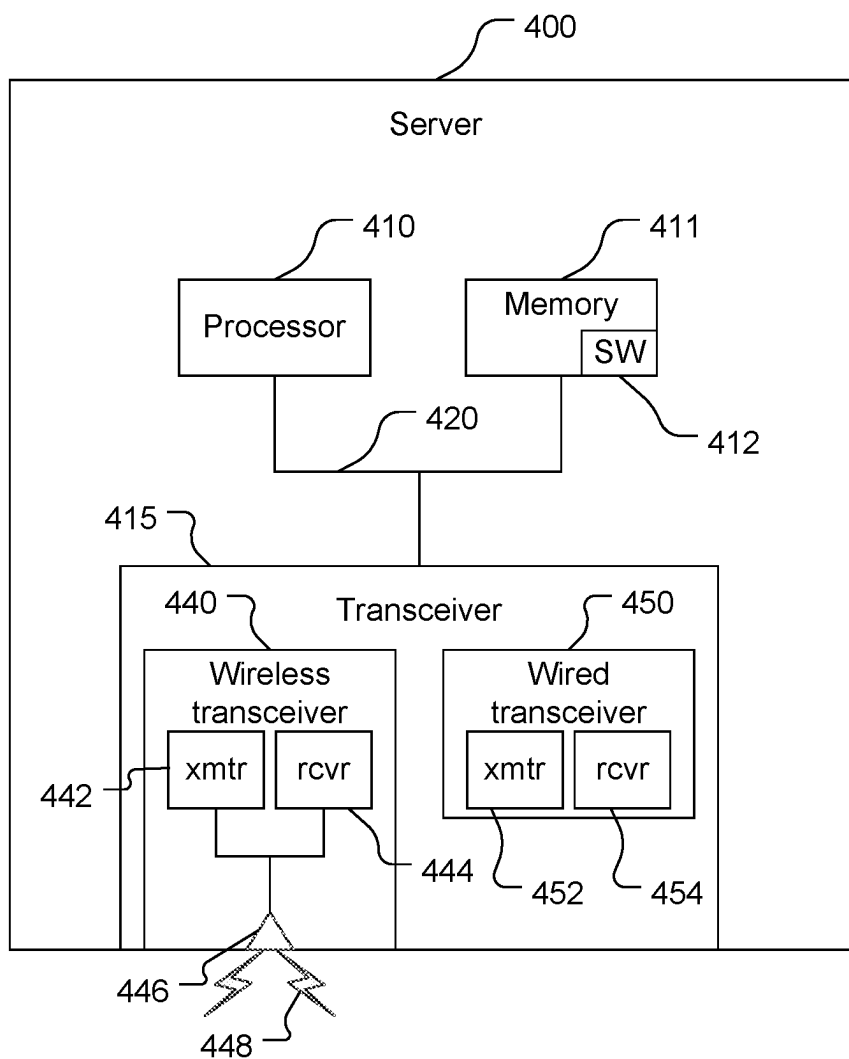


FIG. 4

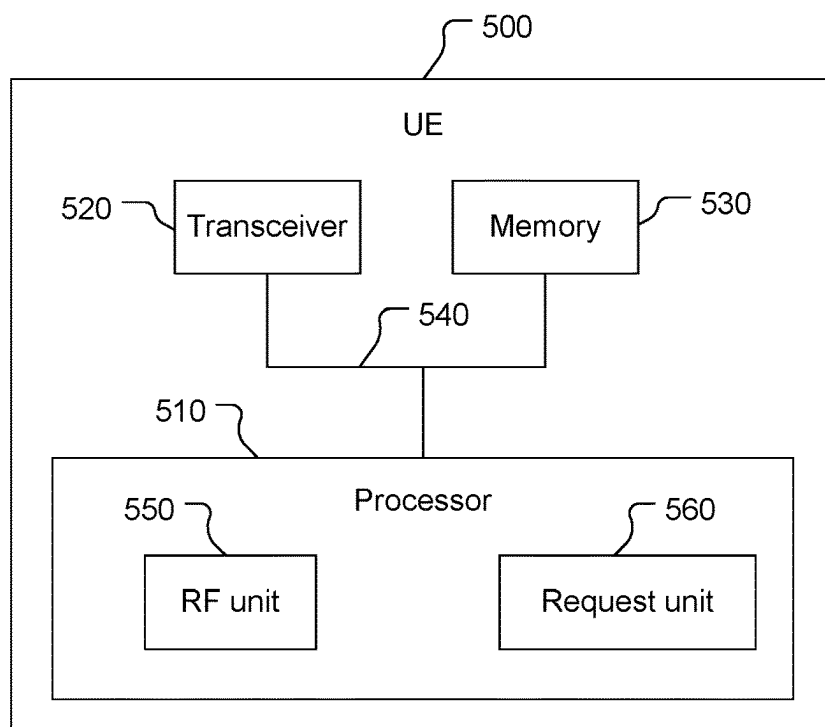


FIG. 5

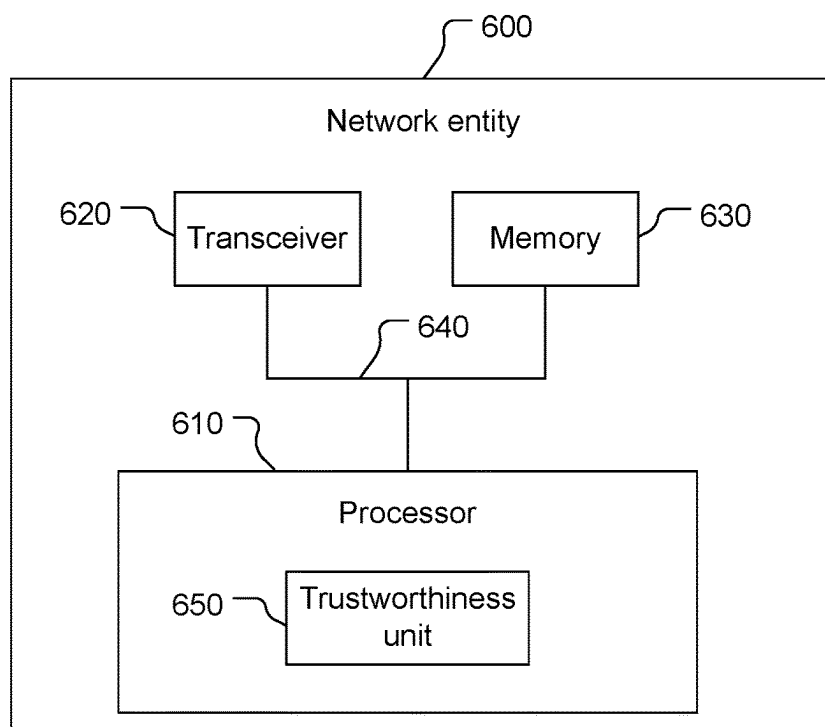


FIG. 6

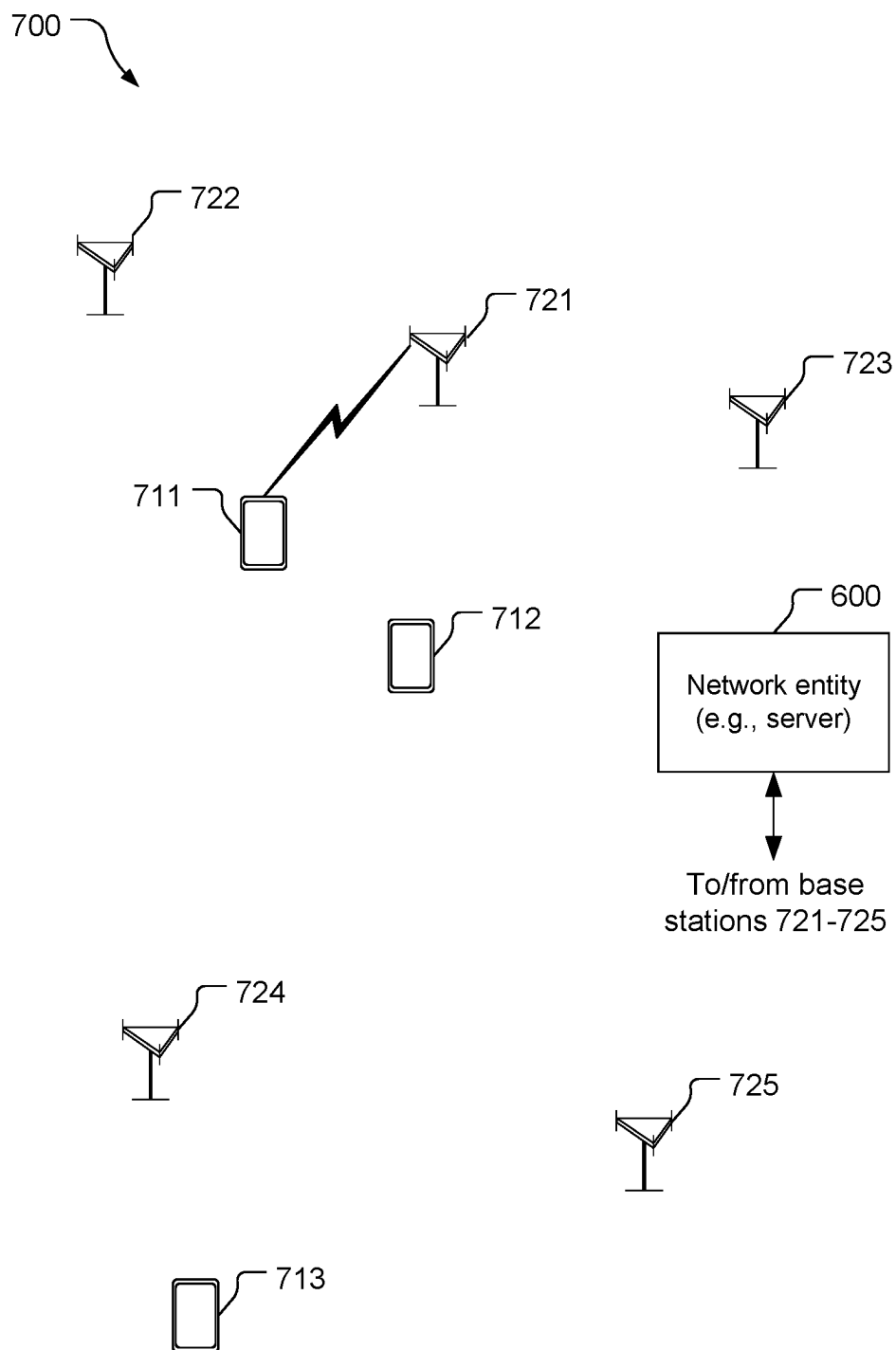


FIG. 7

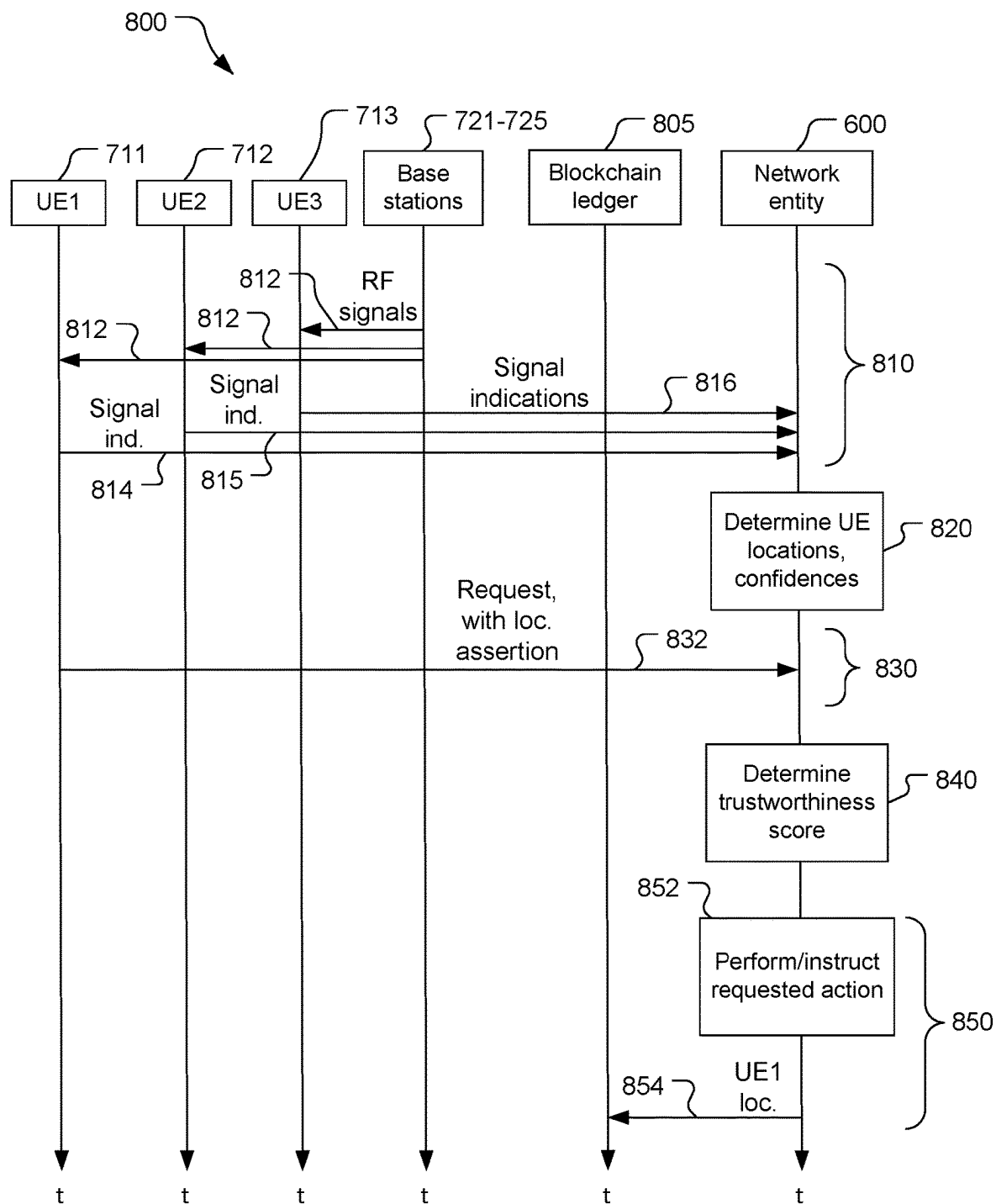


FIG. 8

900

		814	815	816
		UE1	UE2	UE3
921	Serving cell PCID	P1	P1	P4
922	Serving cell ECGI	E1	E1	
923	Neighbor cells PCIDs	P2, P3, P4, P5	P2, P3, P4, P5	P5
924	Neighbor cells ECGIs	E2, E4	E2, E4	
925	SNRs	S1	S4	
926	Doppler estimates	D2, D3, D4, D5		D5
927	RSSIs	R1, R3, R4	R1, R4	R5
928	TDOAs	T2, T3, T4, T5	T2, T3, T4, T5	R5
929	RSRPs	R2, R3, R4, R5	R2, R3, R4, R5	R5
930	RSRQs	R2, R3, R4, R5	R2, R3, R4, R5	R5
931	Frequency domain profiles	F4	F5	

FIG. 9

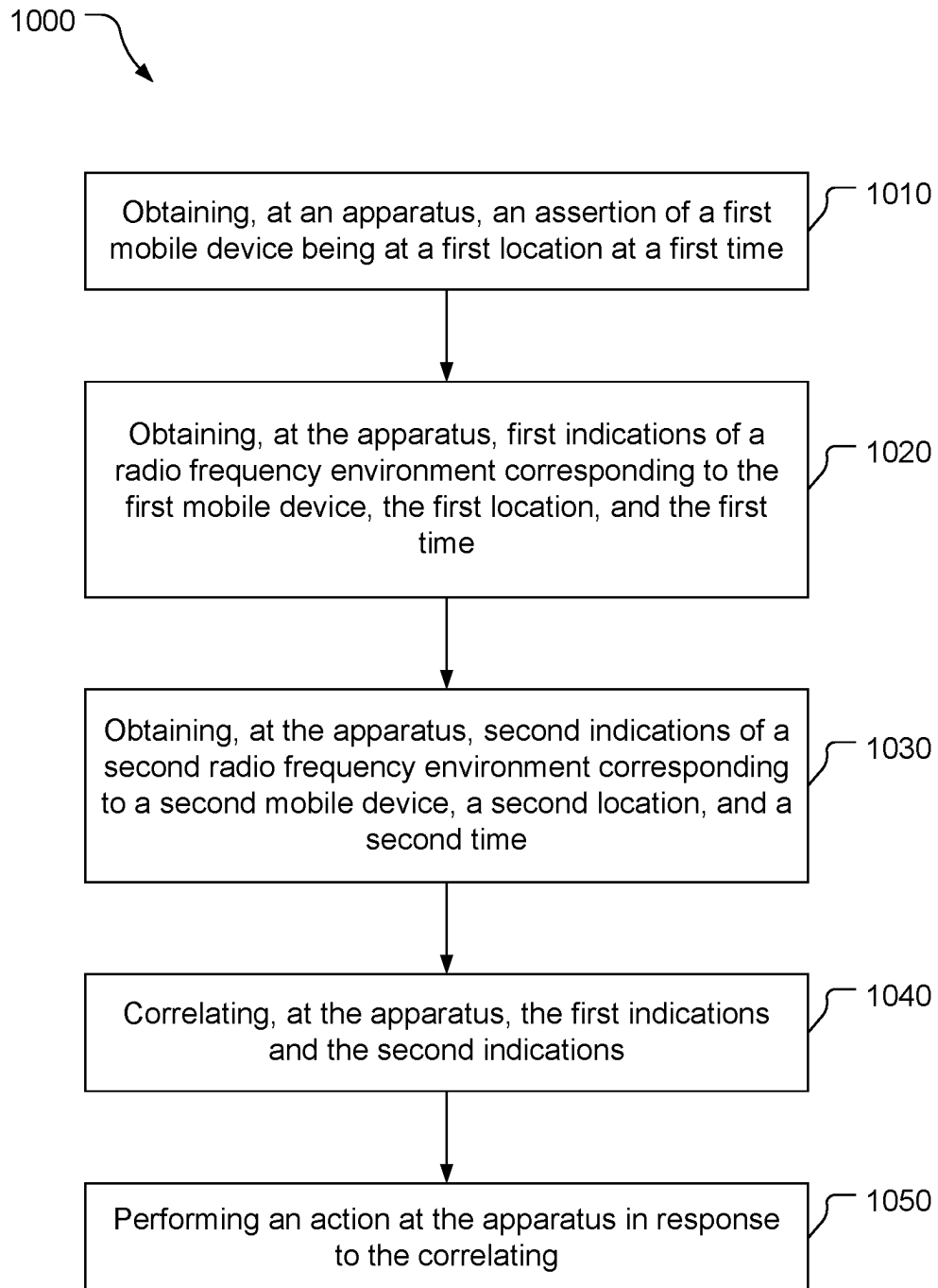


FIG. 10

LOCATION VERIFICATION

BACKGROUND

[0001] Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service, a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax®), a fifth-generation (5G) service (e.g., 5G New Radio (NR)), etc. There are presently many different types of wireless communication systems in use, including Cellular and Personal Communications Service (PCS) systems. Examples of known cellular systems include the cellular Analog Advanced Mobile Phone System (AMPS), and digital cellular systems based on Code Division Multiple Access (CDMA), Frequency Division Multiple Access (FDMA), Orthogonal Frequency Division Multiple Access (OFDMA), Time Division Multiple Access (TDMA), the Global System for Mobile access (GSM) variation of TDMA, etc.

[0002] A fifth generation (5G) mobile standard calls for higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide data rates of several tens of megabits per second to each of tens of thousands of users, with 1 gigabit per second to tens of workers on an office floor. Several hundreds of thousands of simultaneous connections should be supported in order to support large sensor deployments. Consequently, the spectral efficiency of 5G mobile communications should be significantly enhanced compared to the current 4G standard. Furthermore, signaling efficiencies should be enhanced and latency should be substantially reduced compared to current standards.

[0003] It is often desirable to know the location of a user equipment (UE), e.g., a cellular phone, with the terms “location” and “position” being synonymous and used interchangeably herein. A location services (LCS) client may desire to know the location of the UE and may communicate with a location center in order to request the location of the UE. The location center and the UE may exchange messages, as appropriate, to obtain a location estimate for the UE. The location center may return the location estimate to the LCS client, e.g., for use in one or more applications.

[0004] Obtaining the location of a mobile device that is accessing a wireless network may be useful for many applications including, for example, emergency calls, personal navigation, asset tracking, locating a friend or family member, etc. In industrial applications, the location of a mobile device may be necessary for asset tracking, robotic control, and other kinematic operations which may require a precise location of an end effector. Existing positioning methods include methods based on measuring radio signals transmitted from a variety of devices including terrestrial radio sources in a wireless network such as base stations and access points. Stations in a wireless network may be configured to transmit reference signals to enable mobile device to perform positioning measurements.

SUMMARY

[0005] An example mobile device location assertion verification method includes: obtaining, at an apparatus, an

assertion of a first mobile device being at a first location at a first time; obtaining, at the apparatus, first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time; obtaining, at the apparatus, second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time; correlating, at the apparatus, the first indications and the second indications; and performing an action at the apparatus in response to the correlating.

[0006] An example apparatus includes: at least one transceiver; at least one memory; and at least one processor communicatively coupled to the at least one transceiver and the at least one memory and configured to: obtain an assertion of a first mobile device being at a first location at a first time; obtain first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time; obtain second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time; correlate the first indications and the second indications; and perform an action in response to correlation of the first indications and the second indications.

[0007] Another example apparatus includes: means for obtaining an assertion of a first mobile device being at a first location at a first time; means for obtaining first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time; means for obtaining second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time; means for correlating the first indications and the second indications; and means for performing an action in response to correlation of the first indications and the second indications.

[0008] An example non-transitory, processor-readable medium includes processor-readable instructions to cause at least one processor of an apparatus to: obtain an assertion of a first mobile device being at a first location at a first time; obtain first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time; obtain second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time; correlate the first indications and the second indications; and perform an action in response to the correlation of the first indications and the second indications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a simplified diagram of an example wireless communications system.

[0010] FIG. 2 is a block diagram of components of an example user equipment shown in FIG. 1.

[0011] FIG. 3 is a block diagram of components of an example transmission/reception point.

[0012] FIG. 4 is a block diagram of components of a server, various examples of which are shown in FIG. 1.

[0013] FIG. 5 is a simplified block diagram of an example user equipment.

[0014] FIG. 6 is block flow diagram of an example network entity.

[0015] FIG. 7 is a diagram of a wireless signaling system.

[0016] FIG. 8 is a signal and processing flow diagram for verifying a location of a user equipment and performing an action in response thereto.

[0017] FIG. 9 is an example table of signal indications of environmental fingerprints of multiple user equipments.

[0018] FIG. 10 is a block flow diagram of a mobile device location assertion verification method.

DETAILED DESCRIPTION

[0019] Techniques are discussed herein for verifying a location associated with a user equipment. For example, indications of signals received by a first user equipment from base stations may be correlated with indications of signals received by a second user equipment from the base stations. The indications of signals may include signal measurements and/or data extracted from the signals. Correlating the signals may be used to determine a trustworthiness, which may be represented by a determined trustworthiness score, of a location assertion by the first user equipment. The location assertion may be that the first user equipment was at a first location at a first time and the determined trustworthiness represents whether this assertion is accurate, e.g., based on a first fingerprint based on signals received by the first user equipment being similar to a second fingerprint based on signals received by the second user equipment at a second time that is at or near the first time, and with the second user equipment being located at or near the asserted location of the first user equipment. If the asserted location is trustworthy (e.g., is more likely than not to be accurate) for the asserted time, then one or more actions may be taken, e.g., storing the asserted location in association with a blockchain transaction. Other techniques, however, may be used.

[0020] Items and/or techniques described herein may provide one or more of the following capabilities, as well as other capabilities not mentioned. Accuracy of an asserted location of a user equipment may be verified before taking one or more actions. Other capabilities may be provided and not every implementation according to the disclosure must provide any, let alone all, of the capabilities discussed.

[0021] Obtaining the locations of mobile devices that are accessing a wireless network may be useful for many applications including, for example, emergency calls, personal navigation, consumer asset tracking, locating a friend or family member, etc. Existing positioning methods include methods based on measuring radio signals transmitted from a variety of devices or entities including terrestrial radio sources in a wireless network such as base stations and access points. It is expected that standardization for the 5G wireless networks will include support for various positioning methods, which may utilize reference signals transmitted by base stations in a manner similar to which LTE wireless networks currently utilize Positioning Reference Signals (PRS) and/or Cell-specific Reference Signals (CRS) for position determination.

[0022] The description herein may refer to sequences of actions to be performed, for example, by elements of a computing device. Various actions described herein can be performed by specific circuits (e.g., an application specific integrated circuit (ASIC)), by program instructions being executed by one or more processors, or by a combination of both. Sequences of actions described herein may be embodied within a non-transitory computer-readable medium having stored thereon a corresponding set of computer instructions that upon execution would cause an associated processor to perform the functionality described herein. Thus, the various examples described herein may be embod-

ied in a number of different forms, all of which are within the scope of the disclosure, including claimed subject matter.

[0023] As used herein, the terms “user equipment” (UE) and “base station” are not specific to or otherwise limited to any particular Radio Access Technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset tracking device, Internet of Things (IoT) device, automobile, etc.) used to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a Radio Access Network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or UT, a “mobile terminal,” a “mobile station,” a “mobile device,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, WiFi® networks (e.g., based on IEEE (Institute of Electrical and Electronics Engineers) 802.11, etc.) and so on. Two or more UEs may communicate directly in addition to or instead of passing information to each other through a network.

[0024] A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed. Examples of a base station include an Access Point (AP), a Network Node, a NodeB, an evolved NodeB (eNB), or a general Node B (gNodeB, gNB). In addition, in some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions.

[0025] UEs may be embodied by any of a number of types of devices including but not limited to printed circuit (PC) cards, compact flash devices, external or internal modems, wireless or wireline phones, smartphones, tablets, consumer asset tracking devices, asset tags, and so on. A communication link through which UEs can send signals to a RAN is called an uplink channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the RAN can send signals to UEs is called a downlink or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

[0026] As used herein, the term “cell” or “sector” may correspond to one of a plurality of cells of a base station, or to the base station itself, depending on the context. The term “cell” may refer to a logical communication entity used for communication with a base station (for example, over a carrier), and may be associated with an identifier for distinguishing neighboring cells (for example, a physical cell identifier (PCID), a virtual cell identifier (VCID)) operating via the same or a different carrier. In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (for example, machine-type communication (MTC), narrowband Internet-of-Things (NB-IoT), enhanced mobile broadband

(cMBB), or others) that may provide access for different types of devices. In some examples, the term “cell” may refer to a portion of a geographic coverage area (for example, a sector) over which the logical entity operates.

[0027] Referring to FIG. 1, an example of a communication system **100** includes a UE **105**, a UE **106**, a Radio Access Network (RAN), here a Fifth Generation (5G) Next Generation (NG) RAN (NG-RAN) **135**, a 5G Core Network (5GC) **140**, and a server **150**. The UE **105** and/or the UE **106** may be, e.g., an IoT device, a location tracker device, a cellular telephone, a vehicle (e.g., a car, a truck, a bus, a boat, etc.), or another device. A 5G network may also be referred to as a New Radio (NR) network; NG-RAN **135** may be referred to as a 5G RAN or as an NR RAN; and 5GC **140** may be referred to as an NG Core network (NGC). Standardization of an NG-RAN and 5GC is ongoing in the 3rd Generation Partnership Project (3GPP). Accordingly, the NG-RAN **135** and the 5GC **140** may conform to current or future standards for 5G support from 3GPP. The NG-RAN **135** may be another type of RAN, e.g., a 3G RAN, a 4G Long Term Evolution (LTE) RAN, etc. The UE **106** may be configured and coupled similarly to the UE **105** to send and/or receive signals to/from similar other entities in the system **100**, but such signaling is not indicated in FIG. 1 for the sake of simplicity of the figure. Similarly, the discussion focuses on the UE **105** for the sake of simplicity. Additional components of the communication system **100** are described below. The communication system **100** may include additional or alternative components.

[0028] As shown in FIG. 1, the NG-RAN **135** includes NR nodeBs (gNBs) **110a**, **110b**, and a next generation eNodeB (ng-eNB) **114**, and the 5GC **140** includes an Access and Mobility Management Function (AMF) **115**, a Session Management Function (SMF) **117**, a Location Management Function (LMF) **120**, and a Gateway Mobile Location Center (GMLC) **125**. The gNBs **110a**, **110b** and the ng-eNB **114** are communicatively coupled to each other, are each configured to bi-directionally wirelessly communicate with the UE **105**, and are each communicatively coupled to, and configured to bi-directionally communicate with, the AMF **115**. The gNBs **110a**, **110b**, and the ng-eNB **114** may be referred to as base stations (BSs). The AMF **115**, the SMF **117**, the LMF **120**, and the GMLC **125** are communicatively coupled to each other, and the GMLC is communicatively coupled to an external client **130**. The SMF **117** may serve as an initial contact point of a Service Control Function (SCF) (not shown) to create, control, and delete media sessions. Base stations such as the gNBs **110a**, **110b** and/or the ng-eNB **114** may be a macro cell (e.g., a high-power cellular base station), or a small cell (e.g., a low-power cellular base station), or an access point (e.g., a short-range base station configured to communicate with short-range technology such as WiFi®, WiFi®-Direct (WiFi®-D), Bluetooth®, Bluetooth®-low energy (BLE), Zigbee®, etc. One or more base stations, e.g., one or more of the gNBs **110a**, **110b** and/or the ng-eNB **114** may be configured to communicate with the UE **105** via multiple carriers. Each of the gNBs **110a**, **110b** and/or the ng-eNB **114** may provide communication coverage for a respective geographic region, e.g., a cell. Each cell may be partitioned into multiple sectors as a function of the base station antennas.

[0029] FIG. 1 provides a generalized illustration of various components, any or all of which may be utilized as appropriate, and each of which may be duplicated or omitted

as necessary. Specifically, although one UE **105** is illustrated, many UEs (e.g., hundreds, thousands, millions, etc.) may be utilized in the communication system **100**. Similarly, the communication system **100** may include a larger (or smaller) number of SVs (i.e., more or fewer than the four SVs **190-193** shown), gNBs **110a**, **110b**, ng-eNBs **114**, AMFs **115**, external clients **130**, and/or other components. The illustrated connections that connect the various components in the communication system **100** include data and signaling connections which may include additional (intermediary) components, direct or indirect physical and/or wireless connections, and/or additional networks. Furthermore, components may be rearranged, combined, separated, substituted, and/or omitted, depending on desired functionality.

[0030] While FIG. 1 illustrates a 5G-based network, similar network implementations and configurations may be used for other communication technologies, such as 3G, Long Term Evolution (LTE), etc. Implementations described herein (be they for 5G technology and/or for one or more other communication technologies and/or protocols) may be used to transmit (or broadcast) directional synchronization signals, receive and measure directional signals at UEs (e.g., the UE **105**) and/or provide location assistance to the UE **105** (via the GMLC **125** or other location server) and/or compute a location for the UE **105** at a location-capable device such as the UE **105**, the gNB **110a**, **110b**, or the LMF **120** based on measurement quantities received at the UE **105** for such directionally-transmitted signals. The gateway mobile location center (GMLC) **125**, the location management function (LMF) **120**, the access and mobility management function (AMF) **115**, the SMF **117**, the ng-eNB (eNodeB) **114** and the gNBs (gNodeBs) **110a**, **110b** are examples and may be replaced by or include various other location server functionality and/or base station functionality respectively.

[0031] The system **100** is capable of wireless communication in that components of the system **100** can communicate with one another (at least some times using wireless connections) directly or indirectly, e.g., via the gNBs **110a**, **110b**, the ng-eNB **114**, and/or the 5GC **140** (and/or one or more other devices not shown, such as one or more other base transceiver stations). For indirect communications, the communications may be altered during transmission from one entity to another, e.g., to alter header information of data packets, to change format, etc. The UE **105** may include multiple UEs and may be a mobile wireless communication device, but may communicate wirelessly and via wired connections. The UE **105** may be any of a variety of devices, e.g., a smartphone, a tablet computer, a vehicle-based device, etc., but these are examples as the UE **105** is not required to be any of these configurations, and other configurations of UEs may be used. Other UEs may include wearable devices (e.g., smart watches, smart jewelry, smart glasses or headsets, etc.). Still other UEs may be used, whether currently existing or developed in the future. Further, other wireless devices (whether mobile or not) may be implemented within the system **100** and may communicate with each other and/or with the UE **105**, the gNBs **110a**, **110b**, the ng-eNB **114**, the 5GC **140**, and/or the external client **130**. For example, such other devices may include internet of thing (IoT) devices, medical devices, home entertainment and/or automation devices, etc. The 5GC **140** may communicate with the external client **130** (e.g., a

computer system), e.g., to allow the external client **130** to request and/or receive location information regarding the UE **105** (e.g., via the GMLC **125**).

[0032] The UE **105** or other devices may be configured to communicate in various networks and/or for various purposes and/or using various technologies (e.g., 5G, Wi-Fi® communication, multiple frequencies of Wi-Fi® communication, one or more types of communications (e.g., GSM (Global System for Mobiles), CDMA (Code Division Multiple Access), LTE (Long Term Evolution), V2X (Vehicle-to-Everything, e.g., V2P (Vehicle-to-Pedestrian), V2I (Vehicle-to-Infrastructure), V2V (Vehicle-to-Vehicle), etc.), IEEE 802.11p, etc.). V2X communications may be cellular (Cellular-V2X (C-V2X)) and/or Wi-Fi® (e.g., DSRC (Dedicated Short-Range Connection)). The system **100** may support operation on multiple carriers (waveform signals of different frequencies). Multi-carrier transmitters can transmit modulated signals simultaneously on the multiple carriers. Each modulated signal may be a Code Division Multiple Access (CDMA) signal, a Time Division Multiple Access (TDMA) signal, an Orthogonal Frequency Division Multiple Access (OFDMA) signal, a Single-Carrier Frequency Division Multiple Access (SC-FDMA) signal, etc. Each modulated signal may be sent on a different carrier and may carry pilot, overhead information, data, etc. The UEs **105**, **106** may communicate with each other through UE-to-UE sidelink (SL) communications by transmitting over one or more sidelink channels such as a physical sidelink synchronization channel (PSSCH), a physical sidelink broadcast channel (PSBCH), or a physical sidelink control channel (PSCCH). Direct wireless-device-to-wireless-device communications without going through a network may be referred to generally as sidelink communications without limiting the communications to a particular protocol.

[0033] The UE **105** may comprise and/or may be referred to as a device, a mobile device, a wireless device, a mobile terminal, a terminal, a mobile station (MS), a Secure User Plane Location (SUPL) Enabled Terminal (SET), or by some other name. Moreover, the UE **105** may correspond to a cellphone, smartphone, laptop, tablet, PDA, consumer asset tracking device, navigation device, Internet of Things (IoT) device, health monitors, security systems, smart city sensors, smart meters, wearable trackers, or some other portable or moveable device. Typically, though not necessarily, the UE **105** may support wireless communication using one or more Radio Access Technologies (RATs) such as Global System for Mobile communication (GSM), Code Division Multiple Access (CDMA), Wideband CDMA (WCDMA), LTE, High Rate Packet Data (HRPD), IEEE 802.11 Wi-Fi® (also referred to as Wi-Fi®), Bluetooth® (BT), Worldwide Interoperability for Microwave Access (WiMax®), 5G new radio (NR) (e.g., using the NG-RAN **135** and the 5GC **140**), etc. The UE **105** may support wireless communication using a Wireless Local Area Network (WLAN) which may connect to other networks (e.g., the Internet) using a Digital Subscriber Line (DSL) or packet cable, for example. The use of one or more of these RATs may allow the UE **105** to communicate with the external client **130** (e.g., via elements of the 5GC **140** not shown in FIG. 1, or possibly via the GMLC **125**) and/or allow the external client **130** to receive location information regarding the UE **105** (e.g., via the GMLC **125**).

[0034] The UE **105** may include a single entity or may include multiple entities such as in a personal area network

where a user may employ audio, video and/or data I/O (input/output) devices and/or body sensors and a separate wireline or wireless modem. An estimate of a location of the UE **105** may be referred to as a location, location estimate, location fix, fix, position, position estimate, or position fix, and may be geographic, thus providing location coordinates for the UE **105** (e.g., latitude and longitude) which may or may not include an altitude component (e.g., height above sea level, height above or depth below ground level, floor level, or basement level). Alternatively, a location of the UE **105** may be expressed as a civic location (e.g., as a postal address or the designation of some point or small area in a building such as a particular room or floor). A location of the UE **105** may be expressed as an area or volume (defined either geographically or in civic form) within which the UE **105** is expected to be located with some probability or confidence level (e.g., 67%, 95%, etc.). A location of the UE **105** may be expressed as a relative location comprising, for example, a distance and direction from a known location. The relative location may be expressed as relative coordinates (e.g., X, Y (and Z) coordinates) defined relative to some origin at a known location which may be defined, e.g., geographically, in civic terms, or by reference to a point, area, or volume, e.g., indicated on a map, floor plan, or building plan. In the description contained herein, the use of the term location may comprise any of these variants unless indicated otherwise. When computing the location of a UE, it is common to solve for local x, y, and possibly z coordinates and then, if desired, convert the local coordinates into absolute coordinates (e.g., for latitude, longitude, and altitude above or below mean sea level).

[0035] The UE **105** may be configured to communicate with other entities using one or more of a variety of technologies. The UE **105** may be configured to connect indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links. The D2D P2P links may be supported with any appropriate D2D radio access technology (RAT), such as LTE Direct (LTE-D), Wi-Fi® Direct (Wi-Fi®-D), Bluetooth®, and so on. One or more of a group of UEs utilizing D2D communications may be within a geographic coverage area of a Transmission/Reception Point (TRP) such as one or more of the gNBs **110a**, **110b**, and/or the ng-eNB **114**. Other UEs in such a group may be outside such geographic coverage areas, or may be otherwise unable to receive transmissions from a base station. Groups of UEs communicating via D2D communications may utilize a one-to-many (1:M) system in which each UE may transmit to other UEs in the group. A TRP may facilitate scheduling of resources for D2D communications. In other cases, D2D communications may be carried out between UEs without the involvement of a TRP. One or more of a group of UEs utilizing D2D communications may be within a geographic coverage area of a TRP. Other UEs in such a group may be outside such geographic coverage areas, or be otherwise unable to receive transmissions from a base station. Groups of UEs communicating via D2D communications may utilize a one-to-many (1:M) system in which each UE may transmit to other UEs in the group. A TRP may facilitate scheduling of resources for D2D communications. In other cases, D2D communications may be carried out between UEs without the involvement of a TRP.

[0036] Base stations (BSs) in the NG-RAN **135** shown in FIG. 1 include NR Node Bs, referred to as the gNBs **110a**

and 110b. Pairs of the gNBs 110a, 110b in the NG-RAN 135 may be connected to one another via one or more other gNBs. Access to the 5G network is provided to the UE 105 via wireless communication between the UE 105 and one or more of the gNBs 110a, 110b, which may provide wireless communications access to the 5GC 140 on behalf of the UE 105 using 5G. In FIG. 1, the serving gNB for the UE 105 is assumed to be the gNB 110a, although another gNB (e.g., the gNB 110b) may act as a serving gNB if the UE 105 moves to another location or may act as a secondary gNB to provide additional throughput and bandwidth to the UE 105.

[0037] Base stations (BSs) in the NG-RAN 135 shown in FIG. 1 may include the ng-eNB 114, also referred to as a next generation evolved Node B. The ng-eNB 114 may be connected to one or more of the gNBs 110a, 110b in the NG-RAN 135, possibly via one or more other gNBs and/or one or more other ng-eNBs. The ng-eNB 114 may provide LTE wireless access and/or evolved LTE (eLTE) wireless access to the UE 105. One or more of the gNBs 110a, 110b and/or the ng-eNB 114 may be configured to function as positioning-only beacons which may transmit signals to assist with determining the position of the UE 105 but may not receive signals from the UE 105 or from other UEs.

[0038] The gNBs 110a, 110b and/or the ng-eNB 114 may each comprise one or more TRPs. For example, each sector within a cell of a BS may comprise a TRP, although multiple TRPs may share one or more components (e.g., share a processor but have separate antennas). The system 100 may include macro TRPs exclusively or the system 100 may have TRPs of different types, e.g., macro, pico, and/or femto TRPs, etc. A macro TRP may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by terminals with service subscription. A pico TRP may cover a relatively small geographic area (e.g., a pico cell) and may allow unrestricted access by terminals with service subscription. A femto or home TRP may cover a relatively small geographic area (e.g., a femto cell) and may allow restricted access by terminals having association with the femto cell (e.g., terminals for users in a home).

[0039] Each of the gNBs 110a, 110b and/or the ng-eNB 114 may include a radio unit (RU), a distributed unit (DU), and a central unit (CU). For example, the gNB 110b includes an RU 111, a DU 112, and a CU 113. The RU 111, DU 112, and CU 113 divide functionality of the gNB 110b. While the gNB 110b is shown with a single RU, a single DU, and a single CU, a gNB may include one or more RUs, one or more DUs, and/or one or more CUs. An interface between the CU 113 and the DU 112 is referred to as an F1 interface. The RU 111 is configured to perform digital front end (DFE) functions (e.g., analog-to-digital conversion, filtering, power amplification, transmission/reception) and digital beamforming, and includes a portion of the physical (PHY) layer. The RU 111 may perform the DFE using massive multiple input/multiple output (MIMO) and may be integrated with one or more antennas of the gNB 110b. The DU 112 hosts the Radio Link Control (RLC), Medium Access Control (MAC), and physical layers of the gNB 110b. One DU can support one or more cells, and each cell is supported by a single DU. The operation of the DU 112 is controlled by the CU 113. The CU 113 is configured to perform functions for transferring user data, mobility control, radio access network sharing, positioning, session management, etc. although some functions are allocated exclusively to the DU 112. The

CU 113 hosts the Radio Resource Control (RRC), Service Data Adaptation Protocol (SDAP), and Packet Data Convergence Protocol (PDCP) protocols of the gNB 110b. The UE 105 may communicate with the CU 113 via RRC, SDAP, and PDCP layers, with the DU 112 via the RLC, MAC, and PHY layers, and with the RU 111 via the PHY layer.

[0040] As noted, while FIG. 1 depicts nodes configured to communicate according to 5G communication protocols, nodes configured to communicate according to other communication protocols, such as, for example, an LTE protocol or IEEE 802.11x protocol, may be used. For example, in an Evolved Packet System (EPS) providing LTE wireless access to the UE 105, a RAN may comprise an Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN) which may comprise base stations comprising evolved Node Bs (eNBs). A core network for EPS may comprise an Evolved Packet Core (EPC). An EPS may comprise an E-UTRAN plus EPC, where the E-UTRAN corresponds to the NG-RAN 135 and the EPC corresponds to the 5GC 140 in FIG. 1.

[0041] The gNBs 110a, 110b and the ng-eNB 114 may communicate with the AMF 115, which, for positioning functionality, communicates with the LMF 120. The AMF 115 may support mobility of the UE 105, including cell change and handover and may participate in supporting a signaling connection to the UE 105 and possibly data and voice bearers for the UE 105. The LMF 120 may communicate directly with the UE 105, e.g., through wireless communications, or directly with the gNBs 110a, 110b and/or the ng-eNB 114. The LMF 120 may support positioning of the UE 105 when the UE 105 accesses the NG-RAN 135 and may support position procedures/methods such as Observed Time Difference of Arrival (OTDOA) (e.g., Downlink (DL) OTDOA or Uplink (UL) OTDOA), Round Trip Time (RTT), Multi-Cell RTT, Enhanced Cell ID (E-CID), angle of arrival (AoA), angle of departure (AoD), and/or other position methods. The LMF 120 may process location services requests for the UE 105, e.g., received from the AMF 115 or from the GMLC 125. The LMF 120 may be connected to the AMF 115 and/or to the GMLC 125. The LMF 120 may be referred to by other names such as a Location Manager (LM), Location Function (LF), commercial LMF (CLMF), or value added LMF (VLMF). A node/system that implements the LMF 120 may additionally or alternatively implement other types of location-support modules, such as an Enhanced Serving Mobile Location Center (E-SMLC) or a Secure User Plane Location (SUPL) Location Platform (SLP). At least part of the positioning functionality (including derivation of the location of the UE 105) may be performed at the UE 105 (e.g., using signal measurements obtained by the UE 105 for signals transmitted by wireless nodes such as the gNBs 110a, 110b and/or the ng-eNB 114, and/or assistance data provided to the UE 105, e.g., by the LMF 120). The AMF 115 may serve as a control node that processes signaling between the UE 105 and the 5GC 140, and may provide QoS (Quality of Service) flow and session management. The AMF 115 may support mobility of the UE 105 including cell change and handover and may participate in supporting signaling connection to the UE 105.

[0042] The server 150, e.g., a cloud server, is configured to obtain and provide location estimates of the UE 105 to the external client 130. The server 150 may, for example, be configured to run a microservice/service that obtains the

location estimate of the UE 105. The server 150 may, for example, pull the location estimate from (e.g., by sending a location request to) the UE 105, one or more of the gNBs 110a, 110b (e.g., via the RU 111, the DU 112, and the CU 113) and/or the ng-eNB 114, and/or the LMF 120. As another example, the UE 105, one or more of the gNBs 110a, 110b (e.g., via the RU 111, the DU 112, and the CU 113), and/or the LMF 120 may push the location estimate of the UE 105 to the server 150.

[0043] The GMLC 125 may support a location request for the UE 105 received from the external client 130 via the server 150 and may forward such a location request to the AMF 115 for forwarding by the AMF 115 to the LMF 120 or may forward the location request directly to the LMF 120. A location response from the LMF 120 (e.g., containing a location estimate for the UE 105) may be returned to the GMLC 125 either directly or via the AMF 115 and the GMLC 125 may then return the location response (e.g., containing the location estimate) to the external client 130 via the server 150. The GMLC 125 is shown connected to both the AMF 115 and LMF 120, though may not be connected to the AMF 115 or the LMF 120 in some implementations.

[0044] As further illustrated in FIG. 1, the LMF 120 may communicate with the gNBs 110a, 110b and/or the ng-eNB 114 using a New Radio Position Protocol A (which may be referred to as NRPPa or NRPPa), which may be defined in 3GPP Technical Specification (TS) 38.455. NRPPa may be the same as, similar to, or an extension of the LTE Positioning Protocol A (LPPa) defined in 3GPP TS 36.455, with NRPPa messages being transferred between the gNB 110a (or the gNB 110b) and the LMF 120, and/or between the ng-eNB 114 and the LMF 120, via the AMF 115. As further illustrated in FIG. 1, the LMF 120 and the UE 105 may communicate using an LTE Positioning Protocol (LPP), which may be defined in 3GPP TS 36.355. The LMF 120 and the UE 105 may also or instead communicate using a New Radio Positioning Protocol (which may be referred to as NPP or NRPP), which may be the same as, similar to, or an extension of LPP. Here, LPP and/or NPP messages may be transferred between the UE 105 and the LMF 120 via the AMF 115 and the serving gNB 110a, 110b or the serving ng-eNB 114 for the UE 105. For example, LPP and/or NPP messages may be transferred between the LMF 120 and the AMF 115 using a 5G Location Services Application Protocol (LCS AP) and may be transferred between the AMF 115 and the UE 105 using a 5G Non-Access Stratum (NAS) protocol. The LPP and/or NPP protocol may be used to support positioning of the UE 105 using UE-assisted and/or UE-based position methods such as OTDOA and/or E-CID. The NRPPa protocol may be used to support positioning of the UE 105 using network-based position methods such as E-CID (e.g., when used with measurements obtained by the gNB 110a, 110b or the ng-eNB 114) and/or may be used by the LMF 120 to obtain location related information from the gNBs 110a, 110b and/or the ng-eNB 114, such as parameters defining directional SS or PRS transmissions from the gNBs 110a, 110b, and/or the ng-eNB 114. The LMF 120 may be co-located or integrated with a gNB or a TRP, or may be disposed remote from the gNB and/or the TRP and configured to communicate directly or indirectly with the gNB and/or the TRP.

[0045] With a UE-assisted position method, the UE 105 may obtain location measurements and send the measure-

ments to a location server (e.g., the LMF 120) for computation of a location estimate for the UE 105. For example, the location measurements may include one or more of a Received Signal Strength Indication (RSSI), Round Trip signal propagation Time (RTT), Reference Signal Time Difference (RSTD), Reference Signal Received Power (RSRP) and/or Reference Signal Received Quality (RSRQ) for the gNBs 110a, 110b, the ng-eNB 114, and/or a WLAN AP.

[0046] With a UE-based position method, the UE 105 may obtain location measurements (e.g., which may be the same as or similar to location measurements for a UE-assisted position method) and may compute a location of the UE 105 (e.g., with the help of assistance data received from a location server such as the LMF 120 or broadcast by the gNBs 110a, 110b, the ng-eNB 114, or other base stations or APs).

[0047] With a network-based position method, one or more base stations (e.g., the gNBs 110a, 110b, and/or the ng-eNB 114) or APs may obtain location measurements (e.g., measurements of RSSI, RTT, RSRP, RSRQ or Time of Arrival (ToA) for signals transmitted by the UE 105) and/or may receive measurements obtained by the UE 105. The one or more base stations or APs may send the measurements to a location server (e.g., the LMF 120) for computation of a location estimate for the UE 105.

[0048] Information provided by the gNBs 110a, 110b, and/or the ng-eNB 114 to the LMF 120 using NRPPa may include timing and configuration information for directional SS or PRS transmissions and location coordinates. The LMF 120 may provide some or all of this information to the UE 105 as assistance data in an LPP and/or NPP message via the NG-RAN 135 and the 5GC 140.

[0049] An LPP or NPP message sent from the LMF 120 to the UE 105 may instruct the UE 105 to do any of a variety of things depending on desired functionality. For example, the LPP or NPP message could contain an instruction for the UE 105 to obtain measurements for WLAN, E-CID, and/or OTDOA (or some other position method). In the case of E-CID, the LPP or NPP message may instruct the UE 105 to obtain one or more measurement quantities (e.g., beam ID, beam width, mean angle, RSRP, RSRQ measurements) of directional signals transmitted within particular cells supported by one or more of the gNBs 110a, 110b, and/or the ng-eNB 114 (or supported by some other type of base station such as an eNB or WiFi® AP). The UE 105 may send the measurement quantities back to the LMF 120 in an LPP or NPP message (e.g., inside a 5G NAS message) via the serving gNB 110a (or the serving ng-eNB 114) and the AMF 115.

[0050] As noted, while the communication system 100 is described in relation to 5G technology, the communication system 100 may be implemented to support other communication technologies, such as GSM, WCDMA, LTE, etc., that are used for supporting and interacting with mobile devices such as the UE 105 (e.g., to implement voice, data, positioning, and other functionalities). In some such implementations, the 5GC 140 may be configured to control different air interfaces. For example, the 5GC 140 may be connected to a WLAN using a Non-3GPP InterWorking Function (N3IWF, not shown FIG. 1) in the 5GC 140. For example, the WLAN may support IEEE 802.11 WiFi® access for the UE 105 and may comprise one or more WiFi® APs. Here, the N3IWF may connect to the WLAN and to

other elements in the 5GC 140 such as the AMF 115. In some examples, both the NG-RAN 135 and the 5GC 140 may be replaced by one or more other RANs and one or more other core networks. For example, in an EPS, the NG-RAN 135 may be replaced by an E-UTRAN containing eNBs and the 5GC 140 may be replaced by an EPC containing a Mobility Management Entity (MME) in place of the AMF 115, an E-SMLC in place of the LMF 120, and a GMLC that may be similar to the GMLC 125. In such an EPS, the E-SMLC may use LPPa in place of NRPPa to send and receive location information to and from the eNBs in the E-UTRAN and may use LPP to support positioning of the UE 105. In these other examples, positioning of the UE 105 using directional PRSs may be supported in an analogous manner to that described herein for a 5G network with the difference that functions and procedures described herein for the gNBs 110a, 110b, the ng-eNB 114, the AMF 115, and the LMF 120 may, in some cases, apply instead to other network elements such as eNBs, WiFi® APs, an MME, and an E-SMLC.

[0051] As noted, in some examples, positioning functionality may be implemented, at least in part, using the directional SS or PRS beams, sent by base stations (such as the gNBs 110a, 110b, and/or the ng-eNB 114) that are within range of the UE whose position is to be determined (e.g., the UE 105 of FIG. 1). The UE may, in some instances, use the directional SS or PRS beams from a plurality of base stations (such as the gNBs 110a, 110b, the ng-eNB 114, etc.) to compute the position of the UE.

[0052] Referring also to FIG. 2, a UE 200 may be an example of one of the UEs 105, 106 and may comprise a computing platform including a processor 210, memory 211 including software (SW) 212, one or more sensors 213, a transceiver interface 214 for a transceiver 215 (that includes a wireless transceiver 240 and a wired transceiver 250), a user interface 216, a camera 218, and a position device (PD) 219. The processor 210, the memory 211, the sensor(s) 213, the transceiver interface 214, the user interface 216, the camera 218, and the position device 219 may be communicatively coupled to each other by a bus 220 (which may be configured, e.g., for optical and/or electrical communication). One or more of the shown apparatus (e.g., the camera 218, the position device 219, and/or one or more of the sensor(s) 213, etc.) may be omitted from the UE 200. The processor 210 may include one or more hardware devices, e.g., a central processing unit (CPU), a microcontroller, an application specific integrated circuit (ASIC), etc. The processor 210 may comprise multiple processors including a general-purpose/application processor 230, a Digital Signal Processor (DSP) 231, a modem processor 232, a video processor 233, and/or a sensor processor 234. One or more of the processors 230-234 may comprise multiple devices (e.g., multiple processors). For example, the sensor processor 234 may comprise, e.g., processors for RF (radio frequency) sensing (with one or more (cellular) wireless signals transmitted and reflection(s) used to identify, map, and/or track an object), and/or ultrasound, etc. The modem processor 232 may support dual SIM/dual connectivity (or even more SIMs). For example, a SIM (Subscriber Identity Module or Subscriber Identification Module) may be used by an Original Equipment Manufacturer (OEM), and another SIM may be used by an end user of the UE 200 for connectivity. The memory 211 may be a non-transitory storage medium that may include random access memory (RAM), flash

memory, disc memory, and/or read-only memory (ROM), etc. The memory 211 may store the software 212 which may be processor-readable, processor-executable software code containing instructions that may be configured to, when executed, cause the processor 210 to perform various functions described herein. Alternatively, the software 212 may not be directly executable by the processor 210 but may be configured to cause the processor 210, e.g., when compiled and executed, to perform the functions. The description herein may refer to the processor 210 performing a function, but this includes other implementations such as where the processor 210 executes software and/or firmware. The description herein may refer to the processor 210 performing a function as shorthand for one or more of the processors 230-234 performing the function. The description herein may refer to the UE 200 performing a function as shorthand for one or more appropriate components of the UE 200 performing the function. The processor 210 may include a memory with stored instructions in addition to and/or instead of the memory 211. Functionality of the processor 210 is discussed more fully below.

[0053] The configuration of the UE 200 shown in FIG. 2 is an example and not limiting of the disclosure, including the claims, and other configurations may be used. For example, an example configuration of the UE may include one or more of the processors 230-234 of the processor 210, the memory 211, and the wireless transceiver 240. Other example configurations may include one or more of the processors 230-234 of the processor 210, the memory 211, a wireless transceiver, and one or more of the sensor(s) 213, the user interface 216, the camera 218, the PD 219, and/or a wired transceiver.

[0054] The UE 200 may comprise the modem processor 232 that may be capable of performing baseband processing of signals received and down-converted by the transceiver 215. The modem processor 232 may perform baseband processing of signals to be upconverted for transmission by the transceiver 215. Also or alternatively, baseband processing may be performed by the general-purpose/application processor 230 and/or the DSP 231. Other configurations, however, may be used to perform baseband processing.

[0055] The UE 200 may include the sensor(s) 213 that may include, for example, an Inertial Measurement Unit (IMU) 270, one or more magnetometers 271, and/or one or more environment sensors 272. The IMU 270 may comprise, for example, one or more accelerometers 273 (e.g., collectively responding to acceleration of the UE 200 in three dimensions) and/or one or more gyroscopes 274 (e.g., three-dimensional gyroscope(s)). The sensor(s) 213 may include the one or more magnetometers 271 (e.g., three-dimensional magnetometer(s)) to determine orientation (e.g., relative to magnetic north and/or true north) that may be used for any of a variety of purposes, e.g., to support one or more compass applications. The environment sensor(s) 272 may comprise, for example, one or more temperature sensors, one or more barometric pressure sensors, one or more ambient light sensors, one or more camera imagers, and/or one or more microphones, etc. The sensor(s) 213 may generate analog and/or digital signals indications of which may be stored in the memory 211 and processed by the DSP 231 and/or the general-purpose/application processor 230 in support of one or more applications such as, for example, applications directed to positioning and/or navigation operations. The sensor(s) 213 may comprise one or more of other

various types of sensors such as one or more optical sensors, one or more weight sensors, and/or one or more radio frequency (RF) sensors, etc.

[0056] The sensor(s) **213** may be used in relative location measurements, relative location determination, motion determination, etc. Information detected by the sensor(s) **213** may be used for motion detection, relative displacement, dead reckoning, sensor-based location determination, and/or sensor-assisted location determination. The sensor(s) **213** may be useful to determine whether the UE **200** is fixed (stationary) or mobile and/or whether to report certain useful information to the LMF **120** regarding the mobility of the UE **200**. For example, based on the information obtained/measured by the sensor(s) **213**, the UE **200** may notify/report to the LMF **120** that the UE **200** has detected movements or that the UE **200** has moved, and may report the relative displacement/distance (e.g., via dead reckoning, or sensor-based location determination, or sensor-assisted location determination enabled by the sensor(s) **213**). In another example, for relative positioning information, the sensors/IMU may be used to determine the angle and/or orientation of the other device with respect to the UE **200**, etc.

[0057] The IMU **270** may be configured to provide measurements about a direction of motion and/or a speed of motion of the UE **200**, which may be used in relative location determination. For example, the one or more accelerometers **273** and/or the one or more gyroscopes **274** of the IMU **270** may detect, respectively, a linear acceleration and a speed of rotation of the UE **200**. The linear acceleration and speed of rotation measurements of the UE **200** may be integrated over time to determine an instantaneous direction of motion as well as a displacement of the UE **200**. The instantaneous direction of motion and the displacement may be integrated to track a location of the UE **200**. For example, a reference location of the UE **200** may be determined for a moment in time and measurements from the accelerometer(s) **273** and the gyroscope(s) **274** taken after this moment in time may be used in dead reckoning to determine present location of the UE **200** based on movement (direction and distance) of the UE **200** relative to the reference location.

[0058] The magnetometer(s) **271** may determine magnetic field strengths in different directions which may be used to determine orientation of the UE **200**. For example, the orientation may be used to provide a digital compass for the UE **200**. The magnetometer(s) may include a two-dimensional magnetometer configured to detect and provide indications of magnetic field strength in two orthogonal dimensions. The magnetometer(s) **271** may include a three-dimensional magnetometer configured to detect and provide indications of magnetic field strength in three orthogonal dimensions. The magnetometer(s) **271** may provide means for sensing a magnetic field and providing indications of the magnetic field, e.g., to the processor **210**.

[0059] The transceiver **215** may include a wireless transceiver **240** and a wired transceiver **250** configured to communicate with other devices through wireless connections and wired connections, respectively. For example, the wireless transceiver **240** may include a wireless transmitter **242** and a wireless receiver **244** coupled to an antenna **246** for transmitting (e.g., on one or more uplink channels and/or one or more sidelink channels) and/or receiving (e.g., on one or more downlink channels and/or one or more sidelink channels) wireless signals **248** and transducing signals from

the wireless signals **248** to wired (e.g., electrical and/or optical) signals and from wired (e.g., electrical and/or optical) signals to the wireless signals **248**. The wireless transmitter **242** includes appropriate components (e.g., a power amplifier and a digital-to-analog converter). The wireless receiver **244** includes appropriate components (e.g., one or more amplifiers, one or more frequency filters, and an analog-to-digital converter). The wireless transmitter **242** may include multiple transmitters that may be discrete components or combined/integrated components, and/or the wireless receiver **244** may include multiple receivers that may be discrete components or combined/integrated components. The wireless transceiver **240** may be configured to communicate signals (e.g., with TRPs and/or one or more other devices) according to a variety of radio access technologies (RATs) such as 5G New Radio (NR), GSM (Global System for Mobiles), UMTS (Universal Mobile Telecommunications System), AMPS (Advanced Mobile Phone System), CDMA (Code Division Multiple Access), WCDMA (Wideband CDMA), LTE (Long Term Evolution), LTE Direct (LTE-D), 3GPP LTE-V2X (PC5), IEEE 802.11 (including IEEE 802.11p), WiFi®, WiFi® Direct (WiFi®-D), Bluetooth®, Zigbee®, etc. New Radio may use mm-wave frequencies and/or sub-6 GHz frequencies. The wired transceiver **250** may include a wired transmitter **252** and a wired receiver **254** configured for wired communication, e.g., a network interface that may be utilized to communicate with the NG-RAN **135** to send communications to, and receive communications from, the NG-RAN **135**. The wired transmitter **252** may include multiple transmitters that may be discrete components or combined/integrated components, and/or the wired receiver **254** may include multiple receivers that may be discrete components or combined/integrated components. The wired transceiver **250** may be configured, e.g., for optical communication and/or electrical communication. The transceiver **215** may be communicatively coupled to the transceiver interface **214**, e.g., by optical and/or electrical connection. The transceiver interface **214** may be at least partially integrated with the transceiver **215**. The wireless transmitter **242**, the wireless receiver **244**, and/or the antenna **246** may include multiple transmitters, multiple receivers, and/or multiple antennas, respectively, for sending and/or receiving, respectively, appropriate signals.

[0060] The user interface **216** may comprise one or more of several devices such as, for example, a speaker, microphone, display device, vibration device, keyboard, touch screen, etc. The user interface **216** may include more than one of any of these devices. The user interface **216** may be configured to enable a user to interact with one or more applications hosted by the UE **200**. For example, the user interface **216** may store indications of analog and/or digital signals in the memory **211** to be processed by DSP **231** and/or the general-purpose/application processor **230** in response to action from a user. Similarly, applications hosted on the UE **200** may store indications of analog and/or digital signals in the memory **211** to present an output signal to a user. The user interface **216** may include an audio input/output (I/O) device comprising, for example, a speaker, a microphone, digital-to-analog circuitry, analog-to-digital circuitry, an amplifier and/or gain control circuitry (including more than one of any of these devices). Other configurations of an audio I/O device may be used. Also or alternatively, the user interface **216** may comprise one or

more touch sensors responsive to touching and/or pressure, e.g., on a keyboard and/or touch screen of the user interface 216.

[0061] The UE 200 may include the camera 218 for capturing still or moving imagery. The camera 218 may comprise, for example, an imaging sensor (e.g., a charge coupled device or a CMOS (Complementary Metal-Oxide Semiconductor) imager), a lens, analog-to-digital circuitry, frame buffers, etc. Additional processing, conditioning, encoding, and/or compression of signals representing captured images may be performed by the general-purpose/application processor 230 and/or the DSP 231. Also or alternatively, the video processor 233 may perform conditioning, encoding, compression, and/or manipulation of signals representing captured images. The video processor 233 may decode/decompress stored image data for presentation on a display device (not shown), e.g., of the user interface 216.

[0062] The position device (PD) 219 may be configured to determine a position of the UE 200, motion of the UE 200, and/or relative position of the UE 200, and/or time. The PD 219 may work in conjunction with the processor 210 and the memory 211 as appropriate to perform at least a portion of one or more positioning methods, although the description herein may refer to the PD 219 being configured to perform, or performing, in accordance with the positioning method (s). The PD 219 may be configured to determine location of the UE 200 using terrestrial-based signals (e.g., at least some of the wireless signals 248) for trilateration. The PD 219 may be configured to determine location of the UE 200 based on a cell of a serving base station (e.g., a cell center) and/or another technique such as E-CID. The PD 219 may be configured to use one or more images from the camera 218 and image recognition combined with known locations of landmarks (e.g., natural landmarks such as mountains and/or artificial landmarks such as buildings, bridges, streets, etc.) to determine location of the UE 200. The PD 219 may be configured to use one or more other techniques (e.g., relying on the UE's self-reported location (e.g., part of the UE's position beacon)) for determining the location of the UE 200, and may use a combination of techniques to determine the location of the UE 200. The PD 219 may include one or more of the sensors 213 (e.g., gyroscope(s), accelerometer(s), magnetometer(s), etc.) that may sense orientation and/or motion of the UE 200 and provide indications thereof that the processor 210 (e.g., the general-purpose/application processor 230 and/or the DSP 231) may be configured to use to determine motion (e.g., a velocity vector and/or an acceleration vector) of the UE 200. The PD 219 may be configured to provide indications of uncertainty and/or error in the determined position and/or motion. Functionality of the PD 219 may be provided in a variety of manners and/or configurations, e.g., by the general-purpose/application processor 230, the transceiver 215, and/or another component of the UE 200, and may be provided by hardware, software, firmware, or various combinations thereof.

[0063] Referring also to FIG. 3, an example of a TRP 300 of the gNBs 110a, 110b and/or the ng-eNB 114 may comprise a computing platform including a processor 310, memory 311 including software (SW) 312, and a transceiver 315. The processor 310, the memory 311, and the transceiver 315 may be communicatively coupled to each other by a bus 320 (which may be configured, e.g., for optical and/or

electrical communication). One or more of the shown apparatus may be omitted from the TRP 300. The processor 310 may include one or more hardware devices, e.g., a central processing unit (CPU), a microcontroller, an application specific integrated circuit (ASIC), etc. The processor 310 may comprise multiple processors (e.g., including a general-purpose/application processor, a DSP, a modem processor, a video processor, and/or a sensor processor as shown in FIG. 2). The memory 311 may be a non-transitory storage medium that may include random access memory (RAM), flash memory, disc memory, and/or read-only memory (ROM), etc. The memory 311 may store the software 312 which may be processor-readable, processor-executable software code containing instructions that are configured to, when executed, cause the processor 310 to perform various functions described herein. Alternatively, the software 312 may not be directly executable by the processor 310 but may be configured to cause the processor 310, e.g., when compiled and executed, to perform the functions.

[0064] The description herein may refer to the processor 310 performing a function, but this includes other implementations such as where the processor 310 executes software and/or firmware. The description herein may refer to the processor 310 performing a function as shorthand for one or more of the processors contained in the processor 310 performing the function. The description herein may refer to the TRP 300 performing a function as shorthand for one or more appropriate components (e.g., the processor 310 and the memory 311) of the TRP 300 (and thus of one of the gNBs 110a, 110b and/or the ng-eNB 114) performing the function. The processor 310 may include a memory with stored instructions in addition to and/or instead of the memory 311. Functionality of the processor 310 is discussed more fully below.

[0065] The transceiver 315 may include a wireless transceiver 340 and/or a wired transceiver 350 configured to communicate with other devices through wireless connections and wired connections, respectively. For example, the wireless transceiver 340 may include a wireless transmitter 342 and a wireless receiver 344 coupled to one or more antennas 346 for transmitting (e.g., on one or more uplink channels and/or one or more downlink channels) and/or receiving (e.g., on one or more downlink channels and/or one or more uplink channels) wireless signals 348 and transducing signals from the wireless signals 348 to wired (e.g., electrical and/or optical) signals and from wired (e.g., electrical and/or optical) signals to the wireless signals 348. Thus, the wireless transmitter 342 may include multiple transmitters that may be discrete components or combined/integrated components, and/or the wireless receiver 344 may include multiple receivers that may be discrete components or combined/integrated components. The wireless transceiver 340 may be configured to communicate signals (e.g., with the UE 200, one or more other UEs, and/or one or more other devices) according to a variety of radio access technologies (RATs) such as 5G New Radio (NR), GSM (Global System for Mobiles), UMTS (Universal Mobile Telecommunications System), AMPS (Advanced Mobile Phone System), CDMA (Code Division Multiple Access), WCDMA (Wideband CDMA), LTE (Long Term Evolution), LTE Direct (LTE-D), 3GPP LTE-V2X (PC5), IEEE 802.11 (including IEEE 802.11p), WiFi®, WiFi® Direct (WiFi®-D), Bluetooth®, Zigbee®, etc. The wired transceiver 350 may include a wired transmitter 352 and a wired receiver 354

configured for wired communication, e.g., a network interface that may be utilized to communicate with the NG-RAN 135 to send communications to, and receive communications from, the LMF 120, for example, and/or one or more other network entities. The wired transmitter 352 may include multiple transmitters that may be discrete components or combined/integrated components, and/or the wired receiver 354 may include multiple receivers that may be discrete components or combined/integrated components. The wired transceiver 350 may be configured, e.g., for optical communication and/or electrical communication.

[0066] The configuration of the TRP 300 shown in FIG. 3 is an example and not limiting of the disclosure, including the claims, and other configurations may be used. For example, the description herein discusses that the TRP 300 may be configured to perform or performs several functions, but one or more of these functions may be performed by the LMF 120 and/or the UE 200 (i.e., the LMF 120 and/or the UE 200 may be configured to perform one or more of these functions).

[0067] Referring also to FIG. 4, a server 400, of which the LMF 120 may be an example, may comprise a computing platform including a processor 410, memory 411 including software (SW) 412, and a transceiver 415. The processor 410, the memory 411, and the transceiver 415 may be communicatively coupled to each other by a bus 420 (which may be configured, e.g., for optical and/or electrical communication). One or more of the shown apparatus (e.g., a wireless transceiver) may be omitted from the server 400. The processor 410 may include one or more hardware devices, e.g., a central processing unit (CPU), a microcontroller, an application specific integrated circuit (ASIC), etc. The processor 410 may comprise multiple processors (e.g., including a general-purpose/application processor, a DSP, a modem processor, a video processor, and/or a sensor processor as shown in FIG. 2). The memory 411 may be a non-transitory storage medium that may include random access memory (RAM), flash memory, disc memory, and/or read-only memory (ROM), etc. The memory 411 may store the software 412 which may be processor-readable, processor-executable software code containing instructions that are configured to, when executed, cause the processor 410 to perform various functions described herein. Alternatively, the software 412 may not be directly executable by the processor 410 but may be configured to cause the processor 410, e.g., when compiled and executed, to perform the functions. The description herein may refer to the processor 410 performing a function, but this includes other implementations such as where the processor 410 executes software and/or firmware. The description herein may refer to the processor 410 performing a function as shorthand for one or more of the processors contained in the processor 410 performing the function. The description herein may refer to the server 400 performing a function as shorthand for one or more appropriate components of the server 400 performing the function. The processor 410 may include a memory with stored instructions in addition to and/or instead of the memory 411. Functionality of the processor 410 is discussed more fully below.

[0068] The transceiver 415 may include a wireless transceiver 440 and/or a wired transceiver 450 configured to communicate with other devices through wireless connections and wired connections, respectively. For example, the wireless transceiver 440 may include a wireless transmitter

442 and a wireless receiver 444 coupled to one or more antennas 446 for transmitting (e.g., on one or more downlink channels) and/or receiving (e.g., on one or more uplink channels) wireless signals 448 and transducing signals from the wireless signals 448 to wired (e.g., electrical and/or optical) signals and from wired (e.g., electrical and/or optical) signals to the wireless signals 448. Thus, the wireless transmitter 442 may include multiple transmitters that may be discrete components or combined/integrated components, and/or the wireless receiver 444 may include multiple receivers that may be discrete components or combined/integrated components. The wireless transceiver 440 may be configured to communicate signals (e.g., with the UE 200, one or more other UEs, and/or one or more other devices) according to a variety of radio access technologies (RATs) such as 5G New Radio (NR), GSM (Global System for Mobiles), UMTS (Universal Mobile Telecommunications System), AMPS (Advanced Mobile Phone System), CDMA (Code Division Multiple Access), WCDMA (Wideband CDMA), LTE (Long Term Evolution), LTE Direct (LTE-D), 3GPP LTE-V2X (PC5), IEEE 802.11 (including IEEE 802.11p), WiFi®, WiFi® Direct (WiFi®-D), Bluetooth®, Zigbee®, etc. The wired transceiver 450 may include a wired transmitter 452 and a wired receiver 454 configured for wired communication, e.g., a network interface that may be utilized to communicate with the NG-RAN 135 to send communications to, and receive communications from, the TRP 300, for example, and/or one or more other network entities. The wired transmitter 452 may include multiple transmitters that may be discrete components or combined/integrated components, and/or the wired receiver 454 may include multiple receivers that may be discrete components or combined/integrated components. The wired transceiver 450 may be configured, e.g., for optical communication and/or electrical communication.

[0069] The description herein may refer to the processor 410 performing a function, but this includes other implementations such as where the processor 410 executes software (stored in the memory 411) and/or firmware. The description herein may refer to the server 400 performing a function as shorthand for one or more appropriate components (e.g., the processor 410 and the memory 411) of the server 400 performing the function.

[0070] The configuration of the server 400 shown in FIG. 4 is an example and not limiting of the disclosure, including the claims, and other configurations may be used. For example, the wireless transceiver 440 may be omitted. Also or alternatively, the description herein discusses that the server 400 is configured to perform or performs several functions, but one or more of these functions may be performed by the TRP 300 and/or the UE 200 (i.e., the TRP 300 and/or the UE 200 may be configured to perform one or more of these functions).

Positioning Techniques

[0071] For terrestrial positioning of a UE in cellular networks, techniques such as Advanced Forward Link Trilateration (AFLT) and Observed Time Difference Of Arrival (OTDOA) often operate in “UE-assisted” mode in which measurements of reference signals (e.g., PRS, CRS, etc.) transmitted by base stations are taken by the UE and then provided to a location server. The location server calculates the position of the UE based on the measurements and known locations of the base stations. Because these tech-

niques use the location server to calculate the position of the UE, rather than the UE itself, these positioning techniques are not frequently used in applications such as car or cell-phone navigation, which instead typically rely on satellite-based positioning.

[0072] In UE-assisted positioning, the UE sends measurements (e.g., TDOA, Angle of Arrival (AoA), etc.) to the positioning server (e.g., LMF/eSMLC). The positioning server has the base station almanac (BSA) that contains multiple ‘entries’ or ‘records’, one record per cell, where each record contains geographical cell location but also may include other data. An identifier of the ‘record’ among the multiple ‘records’ in the BSA may be referenced. The BSA and the measurements from the UE may be used to compute the position of the UE.

[0073] In conventional UE-based positioning, a UE computes its own position, thus avoiding sending measurements to the network (e.g., location server), which in turn improves latency and scalability. The UE uses relevant BSA record information (e.g., locations of gNBs (more broadly base stations)) from the network. The BSA information may be encrypted. But since the BSA information varies much less often than, for example, the PPP or RTK assistance data described earlier, it may be easier to make the BSA information (compared to the PPP or RTK information) available to UEs that did not subscribe and pay for decryption keys. Transmissions of reference signals by the gNBs make BSA information potentially accessible to crowd-sourcing or war-driving, essentially enabling BSA information to be generated based on in-the-field and/or over-the-top observations.

[0074] Positioning techniques may be characterized and/or assessed based on one or more criteria such as position determination accuracy and/or latency. Latency is a time elapsed between an event that triggers determination of position-related data and the availability of that data at a positioning system interface, e.g., an interface of the LMF **120**. At initialization of a positioning system, the latency for the availability of position-related data is called time to first fix (TTFF), and is larger than latencies after the TTFF. An inverse of a time elapsed between two consecutive position-related data availabilities is called an update rate, i.e., the rate at which position-related data are generated after the first fix. Latency may depend on processing capability, e.g., of the UE. For example, a UE may report a processing capability of the UE as a duration of DL PRS symbols in units of time (e.g., milliseconds) that the UE can process every T amount of time (e.g., T ms) assuming 272 PRB (Physical Resource Block) allocation. Other examples of capabilities that may affect latency are a number of TRPs from which the UE can process PRS, a number of PRS that the UE can process, and a bandwidth of the UE.

[0075] One or more of many different positioning techniques (also called positioning methods) may be used to determine position of an entity such as one of the UEs **105**, **106**. For example, known position-determination techniques include RTT, multi-RTT, OTDOA (also called TDOA and including UL-TDOA and DL-TDOA), Enhanced Cell Identification (E-CID), DL-AOD, UL-AoA, etc. RTT uses a time for a signal to travel from one entity to another and back to determine a range between the two entities. The range, plus a known location of a first one of the entities and an angle between the two entities (e.g., an azimuth angle) can be used to determine a location of the second of the entities. In multi-RTT (also called multi-cell RTT), multiple ranges

from one entity (e.g., a UE) to other entities (e.g., TRPs) and known locations of the other entities may be used to determine the location of the one entity. In TDOA techniques, the difference in travel times between one entity and other entities may be used to determine relative ranges from the other entities and those, combined with known locations of the other entities may be used to determine the location of the one entity. Angles of arrival and/or departure may be used to help determine location of an entity. For example, an angle of arrival or an angle of departure of a signal combined with a range between devices (determined using signal, e.g., a travel time of the signal, a received power of the signal, etc.) and a known location of one of the devices may be used to determine a location of the other device. The angle of arrival or departure may be an azimuth angle relative to a reference direction such as true north. The angle of arrival or departure may be a zenith angle relative to directly upward from an entity (i.e., relative to radially outward from a center of Earth). E-CID uses the identity of a serving cell, the timing advance (i.e., the difference between receive and transmit times at the UE), estimated timing and power of detected neighbor cell signals, and possibly angle of arrival (e.g., of a signal at the UE from the base station or vice versa) to determine location of the UE. In TDOA, the difference in arrival times at a receiving device of signals from different sources along with known locations of the sources and known offset of transmission times from the sources are used to determine the location of the receiving device.

[0076] In a network-centric RTT estimation, the serving base station instructs the UE to scan for/receive RTT measurement signals (e.g., PRS) on serving cells of two or more neighboring base stations (and typically the serving base station, as at least three base stations are needed). The one or more base stations transmit RTT measurement signals on low reuse resources (e.g., resources used by the base station to transmit system information) allocated by the network (e.g., a location server such as the LMF **120**). The UE records the arrival time (also referred to as a receive time, a reception time, a time of reception, or a time of arrival (ToA)) of each RTT measurement signal relative to the UE’s current downlink timing (e.g., as derived by the UE from a DL signal received from its serving base station), and transmits a common or individual RTT response message (e.g., SRS (sounding reference signal) for positioning, i.e., UL-PRS) to the one or more base stations (e.g., when instructed by its serving base station) and may include the time difference TRx-Tx (i.e., UE TRx-Tx or UERx-Tx) between the ToA of the RTT measurement signal and the transmission time of the RTT response message in a payload of each RTT response message. The RTT response message would include a reference signal from which the base station can deduce the ToA of the RTT response. By comparing the difference TTx-Rx between the transmission time of the RTT measurement signal from the base station and the ToA of the RTT response at the base station to the UE-reported time difference TRx-Tx, and subtracting the UERx-Tx, the base station can deduce the propagation time between the base station and the UE, from which the base station can determine the distance between the UE and the base station by assuming the speed of light during this propagation time.

[0077] A UE-centric RTT estimation is similar to the network-based method, except that the UE transmits uplink RTT measurement signal(s) (e.g., when instructed by a

serving base station), which are received by multiple base stations in the neighborhood of the UE. Each involved base station responds with a downlink RTT response message, which may include the time difference between the ToA of the RTT measurement signal at the base station and the transmission time of the RTT response message from the base station in the RTT response message payload.

[0078] For both network-centric and UE-centric procedures, the side (network or UE) that performs the RTT calculation typically (though not always) transmits the first message(s) or signal(s) (e.g., RTT measurement signal(s)), while the other side responds with one or more RTT response message(s) or signal(s) that may include the difference between the ToA of the first message(s) or signal(s) and the transmission time of the RTT response message(s) or signal(s).

[0079] A multi-RTT technique may be used to determine position. For example, a first entity (e.g., a UE) may send out one or more signals (e.g., unicast, multicast, or broadcast from the base station) and multiple second entities (e.g., other TSPs such as base station(s) and/or UE(s)) may receive a signal from the first entity and respond to this received signal. The first entity receives the responses from the multiple second entities. The first entity (or another entity such as an LMF) may use the responses from the second entities to determine ranges to the second entities and may use the multiple ranges and known locations of the second entities to determine the location of the first entity by trilateration.

[0080] In some instances, additional information may be obtained in the form of an angle of arrival (AoA) or angle of departure (AoD) that defines a straight-line direction (e.g., which may be in a horizontal plane or in three dimensions) or possibly a range of directions (e.g., for the UE from the locations of base stations). The intersection of two directions can provide another estimate of the location for the UE.

[0081] For positioning techniques using PRS (Positioning Reference Signal) signals (e.g., TDOA and RTT), PRS signals sent by multiple TRPs are measured and the arrival times of the signals, known transmission times, and known locations of the TRPs used to determine ranges from a UE to the TRPs. For example, an RSTD (Reference Signal Time Difference) may be determined for PRS signals received from multiple TRPs and used in a TDOA technique to determine position (location) of the UE. A positioning reference signal may be referred to as a PRS or a PRS signal. The PRS signals are typically sent using the same power and PRS signals with the same signal characteristics (e.g., same frequency shift) may interfere with each other such that a PRS signal from a more distant TRP may be overwhelmed by a PRS signal from a closer TRP such that the signal from the more distant TRP may not be detected. PRS muting may be used to help reduce interference by muting some PRS signals (reducing the power of the PRS signal, e.g., to zero and thus not transmitting the PRS signal). In this way, a weaker (at the UE) PRS signal may be more easily detected by the UE without a stronger PRS signal interfering with the weaker PRS signal. The term RS, and variations thereof (e.g., PRS, SRS, CSI-RS (Channel State Information-Reference Signal)), may refer to one reference signal or more than one reference signal.

[0082] Positioning reference signals (PRS) include downlink PRS (DL PRS, often referred to simply as PRS) and uplink PRS (UL PRS) (which may be called SRS (Sounding

Reference Signal) for positioning). A PRS may comprise a PN code (pseudorandom number code) or be generated using a PN code (e.g., by modulating a carrier signal with the PN code) such that a source of the PRS may serve as a pseudo-satellite (a pseudolite). The PN code may be unique to the PRS source (at least within a specified area such that identical PRS from different PRS sources do not overlap). PRS may comprise PRS resources and/or PRS resource sets of a frequency layer. A DL PRS positioning frequency layer (or simply a frequency layer) is a collection of DL PRS resource sets, from one or more TRPs, with PRS resource(s) that have common parameters configured by higher-layer parameters DL-PRS-PositioningFrequencyLayer, DL-PRS-ResourceSet, and DL-PRS-Resource. Each frequency layer has a DL PRS subcarrier spacing (SCS) for the DL PRS resource sets and the DL PRS resources in the frequency layer. Each frequency layer has a DL PRS cyclic prefix (CP) for the DL PRS resource sets and the DL PRS resources in the frequency layer. In 5G, a resource block occupies 12 consecutive subcarriers and a specified number of symbols. Common resource blocks are the set of resource blocks that occupy a channel bandwidth. A bandwidth part (BWP) is a set of contiguous common resource blocks and may include all the common resource blocks within a channel bandwidth or a subset of the common resource blocks. Also, a DL PRS Point A parameter defines a frequency of a reference resource block (and the lowest subcarrier of the resource block), with DL PRS resources belonging to the same DL PRS resource set having the same Point A and all DL PRS resource sets belonging to the same frequency layer having the same Point A. A frequency layer also has the same DL PRS bandwidth, the same start PRB (and center frequency), and the same value of comb size (i.e., a frequency of PRS resource elements per symbol such that for comb-N, every Nth resource element is a PRS resource element). A PRS resource set is identified by a PRS resource set ID and may be associated with a particular TRP (identified by a cell ID) transmitted by an antenna panel of a base station. A PRS resource ID in a PRS resource set may be associated with an omnidirectional signal, and/or with a single beam (and/or beam ID) transmitted from a single base station (where a base station may transmit one or more beams). Each PRS resource of a PRS resource set may be transmitted on a different beam and as such, a PRS resource (or simply resource) can also be referred to as a beam. This does not have any implications on whether the base stations and the beams on which PRS are transmitted are known to the UE.

[0083] A TRP may be configured, e.g., by instructions received from a server and/or by software in the TRP, to send DL PRS per a schedule. According to the schedule, the TRP may send the DL PRS intermittently, e.g., periodically at a consistent interval from an initial transmission. The TRP may be configured to send one or more PRS resource sets. A resource set is a collection of PRS resources across one TRP, with the resources having the same periodicity, a common muting pattern configuration (if any), and the same repetition factor across slots. Each of the PRS resource sets comprises multiple PRS resources, with each PRS resource comprising multiple OFDM (Orthogonal Frequency Division Multiplexing) Resource Elements (REs) that may be in multiple Resource Blocks (RBs) within N (one or more) consecutive symbol(s) within a slot. PRS resources (or reference signal (RS) resources generally) may be referred to as OFDM PRS resources (or OFDM RS resources). An

RB is a collection of REs spanning a quantity of one or more consecutive symbols in the time domain and a quantity (12 for a 5G RB) of consecutive sub-carriers in the frequency domain. Each PRS resource is configured with an RE offset, slot offset, a symbol offset within a slot, and a number of consecutive symbols that the PRS resource may occupy within a slot. The RE offset defines the starting RE offset of the first symbol within a DL PRS resource in frequency. The relative RE offsets of the remaining symbols within a DL PRS resource are defined based on the initial offset. The slot offset is the starting slot of the DL PRS resource with respect to a corresponding resource set slot offset. The symbol offset determines the starting symbol of the DL PRS resource within the starting slot. Transmitted REs may repeat across slots, with each transmission being called a repetition such that there may be multiple repetitions in a PRS resource. The DL PRS resources in a DL PRS resource set are associated with the same TRP and each DL PRS resource has a DL PRS resource ID. A DL PRS resource ID in a DL PRS resource set is associated with a single beam transmitted from a single TRP (although a TRP may transmit one or more beams).

[0084] A PRS resource may also be defined by quasi-co-location and start PRB parameters. A quasi-co-location (QCL) parameter may define any quasi-co-location information of the DL PRS resource with other reference signals. The DL PRS may be configured to be QCL type D with a DL PRS or SS/PBCH (Synchronization Signal/Physical Broadcast Channel) Block from a serving cell or a non-serving cell. The DL PRS may be configured to be QCL type C with an SS/PBCH Block from a serving cell or a non-serving cell. The start PRB parameter defines the starting PRB index of the DL PRS resource with respect to reference Point A. The starting PRB index has a granularity of one PRB and may have a minimum value of 0 and a maximum value of 2176 PRBs.

[0085] A PRS resource set is a collection of PRS resources with the same periodicity, same muting pattern configuration (if any), and the same repetition factor across slots. Every time all repetitions of all PRS resources of the PRS resource set are configured to be transmitted is referred to as an “instance”. Therefore, an “instance” of a PRS resource set is a specified number of repetitions for each PRS resource and a specified number of PRS resources within the PRS resource set such that once the specified number of repetitions are transmitted for each of the specified number of PRS resources, the instance is complete. An instance may also be referred to as an “occasion.” A DL PRS configuration including a DL PRS transmission schedule may be provided to a UE to facilitate (or even enable) the UE to measure the DL PRS.

[0086] Multiple frequency layers of PRS may be aggregated to provide an effective bandwidth that is larger than any of the bandwidths of the layers individually. Multiple frequency layers of component carriers (which may be consecutive and/or separate) and meeting criteria such as being quasi co-located (QCLed), and having the same antenna port, may be stitched to provide a larger effective PRS bandwidth (for DL PRS and UL PRS) resulting in increased time of arrival measurement accuracy. Stitching comprises combining PRS measurements over individual bandwidth fragments into a unified piece such that the stitched PRS may be treated as having been taken from a single measurement. Being QCLed, the different frequency layers behave similarly, enabling stitching of the PRS to

yield the larger effective bandwidth. The larger effective bandwidth, which may be referred to as the bandwidth of an aggregated PRS or the frequency bandwidth of an aggregated PRS, provides for better time-domain resolution (e.g., of TDOA). An aggregated PRS includes a collection of PRS resources and each PRS resource of an aggregated PRS may be called a PRS component, and each PRS component may be transmitted on different component carriers, bands, or frequency layers, or on different portions of the same band.

[0087] RTT positioning is an active positioning technique in that RTT uses positioning signals sent by TRPs to UEs and by UEs (that are participating in RTT positioning) to TRPs. The TRPs may send DL-PRS signals that are received by the UEs and the UEs may send SRS (Sounding Reference Signal) signals that are received by multiple TRPs. A sounding reference signal may be referred to as an SRS or an SRS signal. In 5G multi-RTT, coordinated positioning may be used with the UE sending a single UL-SRS for positioning that is received by multiple TRPs instead of sending a separate UL-SRS for positioning for each TRP. A TRP that participates in multi-RTT will typically search for UEs that are currently camped on that TRP (served UEs, with the TRP being a serving TRP) and also UEs that are camped on neighboring TRPs (neighbor UEs). Neighbor TRPs may be TRPs of a single BTS (Base Transceiver Station) (e.g., gNB), or may be a TRP of one BTS and a TRP of a separate BTS. For RTT positioning, including multi-RTT positioning, the DL-PRS signal and the UL-SRS for positioning signal in a PRS/SRS for positioning signal pair used to determine RTT (and thus used to determine range between the UE and the TRP) may occur close in time to each other such that errors due to UE motion and/or UE clock drift and/or TRP clock drift are within acceptable limits. For example, signals in a PRS/SRS for positioning signal pair may be transmitted from the TRP and the UE, respectively, within about 10 ms of each other. With SRS for positioning being sent by UEs, and with PRS and SRS for positioning being conveyed close in time to each other, it has been found that radio-frequency (RF) signal congestion may result (which may cause excessive noise, etc.) especially if many UEs attempt positioning concurrently and/or that computational congestion may result at the TRPs that are trying to measure many UEs concurrently.

[0088] RTT positioning may be UE-based or UE-assisted. In UE-based RTT, the UE 200 determines the RTT and corresponding range to each of the TRPs 300 and the position of the UE 200 based on the ranges to the TRPs 300 and known locations of the TRPs 300. In UE-assisted RTT, the UE 200 measures positioning signals and provides measurement information to the TRP 300, and the TRP 300 determines the RTT and range. The TRP 300 provides ranges to a location server, e.g., the server 400, and the server determines the location of the UE 200, e.g., based on ranges to different TRPs 300. The RTT and/or range may be determined by the TRP 300 that received the signal(s) from the UE 200, by this TRP 300 in combination with one or more other devices, e.g., one or more other TRPs 300 and/or the server 400, or by one or more devices other than the TRP 300 that received the signal(s) from the UE 200.

[0089] Various positioning techniques are supported in 5G NR. The NR native positioning methods supported in 5G NR include DL-only positioning methods, UL-only positioning methods, and DL+UL positioning methods. Downlink-based positioning methods include DL-TDOA and DL-

AoD. Uplink-based positioning methods include UL-TDOA and UL-AoA. Combined DL+UL-based positioning methods include RTT with one base station and RTT with multiple base stations (multi-RTT).

[0090] A position estimate (e.g., for a UE) may be referred to by other names, such as a location estimate, location, position, position fix, fix, or the like. A position estimate may be geodetic and comprise coordinates (e.g., latitude, longitude, and possibly altitude) or may be civic and comprise a street address, postal address, or some other verbal description of a location. A position estimate may further be defined relative to some other known location or defined in absolute terms (e.g., using latitude, longitude, and possibly altitude). A position estimate may include an expected error or uncertainty (e.g., by including an area or volume within which the location is expected to be included with some specified or default level of confidence). Position information may include one or more positioning signal measurements (e.g., of one or more PRS and/or one or more other signals), and/or one or more values (e.g., one or more ranges (possibly including one or more pseudoranges), and/or one or more position estimates, etc.) based on one or more positioning signal measurements.

Asserted Location Verification

[0091] Referring also to FIG. 5, a UE 500 includes a processor 510, a transceiver 520, and a memory 530 communicatively coupled to each other by a bus 540. Even if referred to in the singular, the processor 510 may include one or more processors, the transceiver 520 may include one or more transceivers (e.g., one or more transmitters and/or one or more receivers), and the memory 530 may include one or more memories. The UE 500 may include the components shown in FIG. 5. The UE 500 may include one or more other components such as any of those shown in FIG. 2 such that the UE 200 may be an example of the UE 500. For example, the processor 510 may include one or more of the components of the processor 210. As another example, the UE 500 may be a reduced-capability (RedCap) device such as IoT (Internet of Things) device, e.g., that has limited processing functionality in order to conserve battery power. The transceiver 520 may include one or more of the components of the transceiver 215, e.g., the wireless transmitter 242 and the antenna 246, or the wireless receiver 244 and the antenna 246, or the wireless transmitter 242, the wireless receiver 244, and the antenna 246. Also or alternatively, the transceiver 520 may include the wired transmitter 252 and/or the wired receiver 254. The memory 530 may be configured similarly to the memory 211, e.g., including software with processor-readable instructions configured to cause the processor 510 to perform functions.

[0092] The description herein may refer to the processor 510 performing a function, but this includes other implementations such as where the processor 510 executes software (stored in the memory 530) and/or firmware. The description herein may refer to the UE 500 performing a function as shorthand for one or more appropriate components (e.g., the processor 510 and the memory 530) of the UE 500 performing the function. The processor 510 (possibly in conjunction with the memory 530 and, as appropriate, the transceiver 520) may include an RF unit 550 and a request unit 560. The RF unit 550 may be configured to perform RF signal reception and reporting operations (e.g., receive RF signals, measure and/or decode the RF signals,

and report signal measurements and/or signal contents). The request unit 560 may be configured to transmit explicit and/or implicit requests, e.g., to a network entity, where the requests may be contingent on a location of the UE 500. The RF unit 550 and the request unit 560 are discussed further below, and the description may refer to the processor 510 generally, or the UE 500 generally, as performing any of the functions of the RF unit 550 and/or the request unit 560, with the UE 500 being configured to perform the function(s).

[0093] Referring also to FIG. 6, a network entity 600 includes a processor 610, a transceiver 620, and a memory 630 communicatively coupled to each other by a bus 640. Even if referred to in the singular, the network entity 600 may include one or more network entities, the processor 610 may include one or more processors, the transceiver 620 may include one or more transceivers (e.g., one or more transmitters and/or one or more receivers), and the memory 630 may include one or more memories. The network entity 600 may include the components shown in FIG. 6 and may be configured to be a component of a communication network (e.g., a terrestrial communication network such as a cellular network). The network entity 600 may include one or more other components such as any of those shown in FIG. 4 such that the server 400 may be an example of the network entity 600. For example, the processor 610 may include one or more of the components of the processor 410. The transceiver 620 may include one or more of the components of the transceiver 415. The memory 630 may be configured similarly to the memory 411, e.g., including software with processor-readable instructions configured to cause the processor 610 to perform functions. Also or alternatively, the network entity 600 may include one or more other components such as any of those shown in FIG. 3 such that the TRP 300 may be an example of the network entity 600. For example, the processor 610 may include one or more of the components of the processor 310. The transceiver 620 may include one or more of the components of the transceiver 315. The memory 630 may be configured similarly to the memory 311, e.g., including software with processor-readable instructions configured to cause the processor 610 to perform functions.

[0094] The description herein may refer to the processor 610 performing a function, but this includes other implementations such as where the processor 610 executes software (stored in the memory 630) and/or firmware. The description herein may refer to the network entity 600 performing a function as shorthand for one or more appropriate components (e.g., the processor 610 and the memory 630) of the network entity 600 performing the function. The processor 610 (possibly in conjunction with the memory 630 and, as appropriate, the transceiver 620) may include a trustworthiness unit 650. The trustworthiness unit 650 may be configured to determine a trustworthiness score associated with the UE 500 (e.g., of a location of the UE 500) and to perform or instruct one or more actions based on the location of the UE 500 and the trustworthiness score. The trustworthiness unit 650 is discussed further below, and the description may refer to the processor 610 generally, or the network entity 600 generally, as performing any of the functions of the trustworthiness unit 650, with the network entity 600 being configured to perform the function(s).

[0095] Referring also to FIG. 7, a wireless signaling system 700 includes UEs 711, 712, 713, and base stations 721, 722, 723, 724, 725. The system 700 is an example, and

other configurations of wireless signaling systems (e.g., with other quantities of UEs and/or other quantities of base stations) may be used. Each of the base stations **721-725** may include multiple TRPs, although the discussion herein assumes that each of the UEs **711-713** receives RF signals (excluding multi-path signals) from only one TRP, at most, of each of the base stations **721-725**. Each of the UEs **711-713** may be an example of the UE **500**. The UEs **711-713** may act as RF probes scanning for RF signals (e.g., 4G, 5G, etc.) RF signals from a network that includes the base stations **721-725**. The UEs **711-713** are disposed within a prescribed time and location relationship (e.g., disposed at locations within a threshold amount of time where the locations are within a threshold distance of each other) and may be called a cohort. The threshold distance and/or the threshold time may be use-case dependent. In the example shown, a TRP of the base station **721** is the serving cell for the UE **711** and for the UE **712**. Also in the example shown, the UEs **711, 712** have the same set of neighbor cells, corresponding to TRPs from the base stations **722-725**. In the example shown, a TRP of the base station **724** is a serving cell for the UE **713**, a TRP of the base station **725** is a neighbor cell for the UE **713**, and the base stations **721-723** are out of range (and do not appear as neighbor cells) for the UE **713**. The network entity **600** (e.g., a server) is communicatively coupled (e.g., through wired connections) to the base stations **721-725** and configured for bi-directional communication with the base stations **721-725** (e.g., each of the TRPs of each of the base stations **721-725**).

[0096] Referring also to FIG. **8**, a signal and processing flow **800** for verifying a location of a UE and performing an action based on an asserted location of the UE being trustworthy includes stages shown. The flow **800** is an example flow and not limiting. The flow **800** may be altered, e.g., by having one or more messages and/or one or more stages added, removed, rearranged, combined, performed concurrently, and/or having one or more messages and/or one or more stages split into multiple messages and/or stages. For example, stage **840** may be performed before stage **830**.

[0097] At stage **810**, each of the UEs **711-713** receives RF signals from at least some of the base stations **721-725**, and sends one or more indications corresponding to the received signal(s) to the network entity **600**. For example, the base stations **721-725** may transmit RF signals **812** (with each TRP of each of the base stations **721-725** transmitting one or more respective RF signals). Each of the UEs **711-713** may receive one or more of the RF signals **812**. For example, the UEs **711, 712** may each receive signals from all of the base stations **721-725** and the UE **713** may receive signals from the base stations **724, 725**, but not from the base stations **721-723**. The RF signals may include information for synchronization of the network and for coordinating establishment of data and/or communication sessions, phone calls, positioning sessions, etc. For example, the RF signals **812** may include CRS, PRS, SRS, TRS (Transmission Reference Signals), etc.

[0098] The UEs **711-713** (and/or one or more other devices with a modem), e.g., the RF unit **550** of each of the UEs **711-713**, may be configured to measure and/or extract data from the RF signals **812**, and (compress and) report corresponding signal indications **814, 815, 816** to the network entity **600** (e.g., part of the cloud). For example, the signal indications may be obtained and used to map an area

surrounding each of the UEs **711-713** from the perspective of each of the UEs **711-713**, respectively. The measurements and/or extracted (e.g., decoded) data may be used to determine an RF map of the environment near (e.g., in line of sight (LOS) of) each of the UEs **711-713**. The measurements may include, for example, RSRP, RSRQ, RSSI, TDOA (Time Difference Of Arrival), etc. The extracted data may include signaling information, degraded signal quality information (indicative of multi-path or signal blockage (non-line of sight (NLOS))). The signaling information may include, for each of the RF signals **812**, a code representing the respective signal source (e.g., a respective TRP, cell tower, and/or antenna of a respective one of the base stations **721-725**).

[0099] Referring also to FIG. **9**, the UEs **711-713** may transmit respective signal indications **814, 815, 816** to the network entity **600** indicating measurements and/or extracted information from the respective received signal(s) of the RF signals **812**. A table **900** shows the signal indications **814-816** transmitted by the UEs **711-713** respectively. Each of the UEs **711-713** transmits the respective signal indications **814-816**, and thus the table **900** is a compilation of information transmitted separately in the signal indications **814-816** by the UEs **711-713**. The table **900** includes a serving cell PCID entry **921** (Physical layer Cell ID), a serving cell ECGI entry **922** (E-UTRAN Cell Global Identifier), a neighbor cell PCIDs entry **923**, a neighbor cells ECGIs entry **924**, an SNRs entry **925**, a Doppler estimates entry **926**, an RSSIs entry **927**, a TDOAs entry **928**, an RSRPs entry **929**, an RSRQs entry **930**, and a frequency domain profiles entry **931**. The entries **921-931** in the table **900** are examples, and one or more of the entries **921-931** shown may be omitted and/or one or more other entries for other information included. As shown, not all of the entries **921-931** may have information for any given set of signal indications transmitted by one of the UEs **711-713**. Values in the table **900** use the notation X # where X represents the first letter of the measurement/information type (e.g., “P” for PCID, “D” for Doppler) and # represents the base station, e.g., “4” for the base station **724**. Also, for simplicity of the figure, multiple entries use the same shorthand notation for different measurements and/or for measurements made by different UEs (e.g., R4 for RSSI values measured by the UEs **711, 712**, RSRP values measured by the UEs **711, 712**, and RSRQ values measured by the UEs **711, 712**). These values are different values, but are the respective values corresponding to one or more signals received from the base station **724** at the respective UE **711, 712**. The network entity **600** may anonymize and/or store the information received from the UEs **711-713** in the signal indications **814-816**.

[0100] At stage **820**, the network entity **600** may determine a location for the UE **711** and possibly one or both of the UEs **712, 713**, and may determine a confidence in any of the determined location(s). For example, the processor **610** may retrieve location information (e.g., from respective base station almanacs) for the base stations **721-725** for which signal information is provided in the signal indications **814** (e.g., based on reported cell ID). The processor **610** may perform one or more operations (e.g., Doppler correction and signal processing) to reduce noise and find time-of-arrival peaks. The processor **610** may correlate PCID information and serving cell information to base station almanac information (that has ground truth information for cell

towers) to determine UE position. The processor 610 may use the base station location information and one or more signal measurements (e.g., time of arrival, received signal power, etc.) to determine distances to different base stations 721-725 to perform trilateration to determine the location of the UE 711. The same process may be performed (if sufficient information is available) to determine the location of the UE 712 and/or the location of the UE 713. The network entity 600 may determine a confidence in any determined location. The confidence may be expressed as a CEP-style (Circular Error Probable style) value (e.g., X % sure that a UE is within N meters of a determined/asserted latitude and longitude location). The network entity 600 may identify RF environments indicated by the signal indications 814-816 and associate the respective environment with the respective UE location. The network entity 600 may track the UE locations over time.

[0101] At stage 830, the UE 711 may transmit, to the network entity 600, a request 832 for one or more operations, with the request 832 including an assertion (explicitly and/or implicitly) of the location of the UE 711. For example, the request 832 may include a latitude and longitude for the UE 711. As another example, the request 832 may be for one or more operations that require the UE 711 to be at a particular location (e.g., a particular region), and thus the request 832 implies that the UE 711 is at the particular location. As another example, the request 832 may indicate that the UE 711 is at the same location as the UE 712, and thus imply that the latitude and longitude of the UE 711 are the same location as the latitude and longitude of the UE 712. As another example, the assertion may be in the form of one or more signal measurements such that the asserted location is the location derivable from the measurement(s) and the confidence of the derived location. A requested operation may be, for example, to store the location of the UE 711 in association with a transaction (e.g., a change of possession and/or ownership of one or more items) as part of a blockchain.

[0102] At stage 840, the network entity 600, e.g., the trustworthiness unit 650, may determine a trustworthiness score for the request 832, e.g., for the assertion of the location of the UE 711 for a corresponding time. For example, the network entity 600 may use information from the UE 712 (and/or one or more other UEs) to corroborate information from the UE 711. The more devices from which the network entity 600 uses information to corroborate the information from the UE 711, the more reliable the determined trustworthiness score may be. The trustworthiness score provides an indication of trustworthiness of a determined or asserted location (for a corresponding time (e.g., time asserted explicitly or implicitly (e.g., time of measurements used to determine the location)) and confidence. The trustworthiness unit 650 may analyze one or more factors to determine the trustworthiness score. The factors may be weighted, giving more influence on the trustworthiness score to one or more factors than to one or more other factors. Initial weightings may be determined heuristically, and weightings thereafter may be determined using one or more machine learning techniques. To determine the trustworthiness score, the trustworthiness unit 650 may, for example, corroborate fingerprints from different UEs. The fingerprints represent slices of time from a device's (e.g., a UE's) perspective. Fingerprints from multiple UEs may be used by comparing similar fingerprint portions to add or

subtract credence to an assertion of a UE being at a particular location at a particular time. For example, if the UE 711 asserts that the UE 711 was at a particular location at a particular time, and the UE 712 observed a similar environment (e.g., more than 50%, or more than 80%, or more than 90%) of the same (RF) fingerprint, and the UE 712 was near the particular location, then the UE 711 was likely at the particular location, and the determined trustworthiness score will reflect this (e.g., be high, such as near 1.0). By comparing fingerprints of more UEs, the trustworthiness score may be more accurate/reliable, whether that indicates that the UE 711 was at the asserted location or not. The comparison of fingerprints may be called cohort verification. To determine the fingerprints, the trustworthiness unit 650 may consider, for example, serving cells of different UEs, commonly-observed cells by different UEs, signal measurements (e.g., SNR, RSSI, TDOA, RSRP, etc.) of commonly-observed signals by different UEs, Doppler estimates of commonly-observed cells, frequency domain profiles of commonly-observed signals, time domain impulse responses of commonly-observed signals, and/or model sizes of analyses of commonly-observed signals.

[0103] The trustworthiness unit 650 may consider the serving cells of different UEs in determining the fingerprint and in turn the trustworthiness score. This factor may be the most-heavily-weighted factor considered, because different UEs having the same serving cell is a very strong indicator that the UEs are located in close proximity to each other. This factor becomes an even stronger indicator of close proximity if one or more neighbor cells are also shared by the UEs.

[0104] The trustworthiness unit 650 may consider commonly-observed cells, and whether the observed cell is a serving cell or not, in determining the fingerprint and in turn the trustworthiness score. If different UEs observe the same cell, then this is a good indicator that the UEs are located close to each other (or at least locations corresponding to times of the respective observations are close to each other). Thus, commonly-observed cells will increase the trustworthiness score. The commonly-observed cell determination may be based on receiving signals with the same PCID and/or the same EGCI. Different UEs may extract and report PCIDs (which are not globally unique) from signals received, and the network entity 600 may determine ECIGIs (which are globally unique) from the PCIDs, respective base station almanacs, and the determined location(s) of the reporting UEs or a determined location of the cohort. If there are multiple cells that are commonly observed, then this may increase the trustworthiness score, and may do so in a non-linear manner (with each additional commonly-observed cell increasing the trustworthiness score by more than a linear increase from N-1 commonly-observed cells to N commonly-observed cells). There are 504 possible PCIDs, and thus if different UEs each observe the same ten (10) PCIDs, then the probability of this being coincidental (assuming that PCIDs are uniformly distributed) would be about 1 in 266391331458861886200. Consequently, if ten cells were commonly observed, then the trustworthiness score should reflect near certainty of close location of the two UEs. Further, if the same ECIGI (which is globally unique) is observed by different UEs, then this is a very strong indication of the UEs being closely located and thus in the same cohort.

[0105] The trustworthiness unit **650** may consider one or more signal measurements in determining the fingerprint and in turn the trustworthiness score. For example, the trustworthiness unit **650** may determine an SNR of each detection (e.g., each measurement of one of the RF signals **812**). In many cases, the SNR of a given detection is related to the distance between a UE and the transmitting base station. If the environment around the UE is dense urban, then multi-path and shadowing from obstructions may modify this relationship drastically. Similarly, the SNR (RSRQ equivalent) and the relative signal level may be used as a proxy rough estimate of the distance between a receiving UE and the transmitting base station. As another example, the trustworthiness unit **650** may consider the RSSI of each detection in determining the fingerprint for a UE. For example, RSRP (determined from cell transmission of a reference sequence) may be used to estimate location of a UE, e.g., with an algorithm that uses ECID that uses Timing Advance. As another example, the RSRP of individual cells (e.g., sectors) may not be available, in which case the RSRP of the servicing cell and/or an auxiliary cell may be used. Using frequency domain correlation coefficients, the relative received power of a reference signal from each sector may be estimated. This estimated power is the RSSI, and the network entity **600** may relate the RSSI of the serving cell to the RSRP of the serving cell, and use this relationship to estimate the RSRP of other detections. These RSRPs may be used to determine the location of the receiving UE. As another example, the TDOA of signals received from the same cell(s) may be used to determine a fingerprint. The trustworthiness unit **650** may use the TDOA of signals received by a UE and the RTTD (Round Trip Time Difference) of the serving cell to estimate the location of the UE. The relative delays between the UEs **711**, **712** and the base stations **721-725** for commonly-received ones of the RF signals **812** may be compared, e.g., by the trustworthiness unit **650**. The more similar the relative delays, the stronger the trustworthiness score will be. The TDOA values may include accumulated data from dwell aggregation and possibly from tracking algorithms that provide smoothing and averaging of historical data.

[0106] The trustworthiness unit **650** may consider Doppler estimates for detections in determining the fingerprint and in turn the trustworthiness score. If a UE is stationary, and the CFO (Carrier Frequency Offset) for the UE has been correctly estimated, then there should be little or no residual Doppler seen for other cells detected. The trustworthiness unit **650** may compare the Doppler offsets seen by different UEs, e.g., the UEs **711**, **712**, to determine whether the UEs are moving together, with similar Doppler offsets resulting in a higher trustworthiness score.

[0107] The trustworthiness unit **650** may consider frequency domain profiles for detections in determining the fingerprint and in turn the trustworthiness score. Frequency domain correlation coefficients indicate the possible existence of multi-path components of a channel through which a signal has travelled. The trustworthiness unit **650** may compare the frequency domain correlation coefficients from different UEs, with similar the trustworthiness score being increased if the frequency domain correlation coefficients from different UEs are similar (e.g., have more than a threshold level of similarity, e.g., have an average difference below a threshold).

[0108] The trustworthiness unit **650** may consider time domain impulse responses in determining the fingerprint and in turn the trustworthiness score. The trustworthiness unit **650** may compare time domain impulse responses of detections of the same signals by multiple UEs (i.e., commonly-detected signals). For example, in addition to or instead of TDOA, a multi-path characteristic may be observed in a time domain impulse response. This multi-path characteristic is a fingerprint (or a portion thereof) of a channel through which a signal has passed. The multi-path characteristics from detections from multiple UEs may be compared (e.g., for common PCIDs), with more similarity between the multi-path characteristics increasing the trustworthiness score.

[0109] The trustworthiness unit **650** may consider analysis model size in determining the fingerprint and in turn the trustworthiness score. For example, the trustworthiness unit **650** may consider model size of an ECID UE position solution that uses Timing Advance as a primary feature and/or model size of an OTDOA UE position solution. The OTDOA solution may assess the complexity of a multi-path channel, including a model size of the number of distinct delays in a multi-path response. The trustworthiness unit **650** may compare the model size for each OTDOA solution estimate for PCIDs of commonly-observed signals received by the UEs **711**, **712**. The trustworthiness score may be directly correlated with the model size, such that the trustworthiness score is increased as the model size increases, and the trustworthiness score is decreased as the model size decreases.

[0110] The trustworthiness score provides a weighted combination of statistics measured in the fingerprinting analysis of the UEs within a cohort, e.g., the UEs **711**, **712**, **713**. The trustworthiness score may be used to assess the accuracy of the determined and/or asserted location of the UE **711** and the likelihood that the UE **711** is at this location. Whether the UEs **711**, **712** have the same serving cell and what other cells the UEs **711**, **712** both observe may be more indicative of the UEs **711**, **712** being located near each other than other fingerprint measures, and thus these factors may be weighted more heavily in determining the trustworthiness score than other fingerprint measures. The confidence in a determined location and/or confidences of base station almanac locations may be used as weights in determining the trustworthiness score. For example, a base station almanac will include a location estimate of the base station, and a latitude/longitude location may be weighted more heavily than a street address as the location as the latitude/longitude location may have a higher confidence than the street address due to being more precise.

[0111] At stage **850**, one or more actions may be taken, or avoided, based on the trustworthiness score determined at stage **840** and the request received at stage **830**. For example, having performed cohort verification, at sub-stage **852** the network entity **650** may use the result of the verification to prompt performance of one or more requested actions (based on the trustworthiness score indicating that the location of the UE **711** is trustworthy, e.g., above a threshold such as 50% (or 60%, or 70%, or 80%, or 90%) likely to be accurate for the time corresponding to the asserted location) or to avoid performance of one or more actions (based on the trustworthiness score indicating that the location of the UE **711** is not trustworthy, e.g., below the threshold of likely accuracy). For example, the network

entity **650** may transmit a location message **854** containing an indication of a location of the UE **711** to a blockchain ledger **805** for storage in the blockchain ledger **805** in association with one or more transactions (e.g., a change of ownership and/or possession of one or more items associated with the UE **711**). This may provide proof of location associated with a change of custody, e.g., of a good from one point in a supply chain to another point in the supply chain.

[0112] Referring to FIG. 10, with further reference to FIGS. 1-9, a mobile device location assertion verification method **1000** includes the stages shown. The method **1000** is, however, an example only and not limiting. The method **1000** may be altered, e.g., by having one or more stages added, removed, rearranged, combined, performed concurrently, and/or by having one or more single stages split into multiple stages.

[0113] At stage **1010**, the method **1000** includes obtaining, at an apparatus, an assertion of a first mobile device being at a first location at a first time. For example, at stage **830**, the network entity **600** receives the request **832** from the UE **711**. The processor **610**, possibly in combination with the memory **630**, in combination with the transceiver **620** (e.g., a wired receiver and/or a wireless receiver and an antenna) may comprise means for obtaining the assertion. As another example, one portion of the processor **510** may receive the request from another portion of the processor **510**, e.g., explicitly or implicitly (as one or more signal measurements). The processor **510**, possibly in combination with the memory **530**, may comprise means for obtaining the assertion.

[0114] At stage **1020**, the method **1000** includes obtaining, at the apparatus, first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time. For example, at stage **810**, the network entity **650** may receive the signal indications **814** from the UE **711**. The processor **610**, possibly in combination with the memory **630**, in combination with the transceiver **620** (e.g., a wired receiver and/or a wireless receiver and an antenna) may comprise means for obtaining the first indications. As another example, one portion of the processor **510** may receive the first indications from another portion of the processor **510**. The processor **510**, possibly in combination with the memory **530**, may comprise means for obtaining the first indications.

[0115] At stage **1030**, the method **1000** includes obtaining, at the apparatus, second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time. For example, at stage **810**, the network entity **650** may receive the signal indications **815** from the UE **712**. The processor **610**, possibly in combination with the memory **630**, in combination with the transceiver **620** (e.g., a wired receiver and/or a wireless receiver and an antenna) may comprise means for obtaining the second indications. As another example, one portion of the processor **510** may receive the second indications from another portion of the processor **510**. The processor **510**, possibly in combination with the memory **530**, may comprise means for obtaining the second indications.

[0116] At stage **1040**, the method **1000** includes correlating, at the apparatus, the first indications and the second indications. For example, at stage **840**, the network entity **650** may correlate the signal indications **814**, **815**, or at least similar portions thereof (e.g., similar signal measurements).

The processor **610**, possibly in combination with the memory **630**, may comprise means for correlating the first indications and the second indications. As another example, the UE **711** may receive the second indications (e.g., from the UE **712** via the transceiver **520** (e.g., a wireless receiver and an antenna) of the UE **711** and/or from the network entity **650** via the transceiver **520** (e.g., a wireless receiver and an antenna) of the UE **711**) and correlate the first indications and the second indications. The processor **510**, possibly in combination with the memory **530**, in combination with the transceiver **520** may comprise means for correlating the first indications and the second indications.

[0117] At stage **1050**, the method **1000** includes performing an action at the apparatus in response to the correlating. For example, at stage **850**, the network entity **650** may perform an action, such as transmitting the message **854** to the blockchain ledger **805** such that the location of the UE **711** may be stored in the blockchain ledger **805** in association with a transaction. The processor **610**, possibly in combination with the memory **630**, possibly in combination with the transceiver **620** (e.g., a wired transmitter and/or a wireless transmitter and an antenna), may comprise means for performing the action. As another example, the UE **711** may perform an action, such as transmitting the message **854** (or a similar message) to the blockchain ledger **805** such that the location of the UE **711** may be stored in the blockchain ledger **805** in association with a transaction. The processor **510**, possibly in combination with the memory **530**, possibly in combination with the transceiver **520** (e.g., a wired transmitter and/or a wireless transmitter and an antenna), may comprise means for performing the action.

[0118] Implementations of the method **1000** may include one or more of the following features. In an example implementation, performing the action comprises performing the action in response to the correlating indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time. For example, performing the action may be based on the mobile device being more likely than not to have been within the threshold distance of the first location within the threshold time of the first time. As another example, performing the action at the apparatus may be in response to the correlating indicating that the first mobile device is greater than another level of likelihood (e.g., 60%, 70%, 80%, or 90%) that the first mobile device was within the threshold distance of the first location within the threshold time of the first time. The second location may be required to be within the threshold distance of the first location and the second location may be required to be within the threshold time of the first time. The threshold distance and/or the threshold time may be use-case dependent, and may be determined/tuned based on heuristic and ML/AI modeling (machine learning/artificial intelligence modeling). Examples of time and distance thresholds are 12 hours and 2,000 m. In a further example implementation, performing the action comprises storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time. For example, at stage **850**, the network entity **650** may transmit the message **854** to the blockchain ledger **805** such that the location of the UE **711** may be stored in the blockchain ledger **805** in association with a transaction. The processor **610**, possibly in combination

with the memory **630**, possibly in combination with the transceiver **620** (e.g., a wired transmitter and/or a wireless transmitter and an antenna), may comprise means for storing the location verification indication. As another example, the UE **711** may transmit the message **854** (or a similar message) to the blockchain ledger **805** such that the location of the UE **711** may be stored in the blockchain ledger **805** in association with a transaction. The processor **510**, possibly in combination with the memory **530**, possibly in combination with the transceiver **520** (e.g., a wired transmitter and/or a wireless transmitter and an antenna), may comprise means for transmitting the location verification indication. The location verification indication may indicate that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time by indicating that the first mobile device was at the first location at the first time.

[0119] Also or alternatively, implementations of the method **1000** may include one or more of the following features. In an example implementation, the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time, and the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time. For example, the signal indications **814** may include a PCID (or ECGI) of a serving cell for the UE **711**, e.g., as shown in the entries **921**, **922** in the table **900**, and the signal indications **815** may include a PCID (or ECGI) of a serving cell for the UE **712**, e.g., as shown in the entries **921**, **922** in the table **900**. In a further example implementation, the correlating comprises determining whether both the first indications and the second indications include a common globally-unique cell identifier. For example, the processor **610**, e.g., the trustworthiness unit **650**, or the processor **510**, may determine whether the signal indications **814**, **815** include an ECGI for the respective serving cells for the UEs **711**, **712**.

[0120] Also or alternatively, implementations of the method **1000** may include one or more of the following features. In an example implementation, the correlating comprises comparing a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source. In another example implementation, obtaining the first indications comprises receiving the first indications at the apparatus from the first mobile device, and the method **1000** further includes: determining a latitude and longitude of the first mobile device based on at least some of the first indications; determining a confidence of the latitude and longitude; and determining a trustworthiness score based on the correlating, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude. For example, at stage **820**, the network entity **650** may determine a latitude and longitude of the UE **711** and a confidence thereof. At stage **850**, the network entity **650** may determine the trustworthiness score based on correlating at least some of the

signal indications **814**, **815**. The processor **610**, possibly in combination with the memory **630**, may comprise means for determining the latitude, the longitude, the confidence, and the trustworthiness score. As another example, the processor **510**, possibly in combination with the memory **530**, may comprise means for determining the latitude, the longitude, the confidence, and the trustworthiness score.

Implementation Examples

[0121] Implementation examples are provided in the following numbered clauses.

[0122] Clause 1. A mobile device location assertion verification method comprising:

[0123] obtaining, at an apparatus, an assertion of a first mobile device being at a first location at a first time;

[0124] obtaining, at the apparatus, first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;

[0125] obtaining, at the apparatus, second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time;

[0126] correlating, at the apparatus, the first indications and the second indications; and

[0127] performing an action at the apparatus in response to the correlating.

[0128] Clause 2. The method of clause 1, wherein performing the action comprises performing the action in response to the correlating indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.

[0129] Clause 3. The method of clause 2, wherein performing the action comprises storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.

[0130] Clause 4. The method of clause 1, wherein:

[0131] the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and

[0132] the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.

[0133] Clause 5. The method of clause 4, wherein the correlating comprises determining whether both the first indications and the second indications include a common globally-unique cell identifier.

[0134] Clause 6. The method of clause 1, wherein the correlating comprises comparing a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.

[0135] Clause 7. The method of clause 1, wherein obtaining the first indications comprises receiving the

- first indications at the apparatus from the first mobile device, the method further comprising:
- [0136] determining a latitude and longitude of the first mobile device based on at least some of the first indications;
 - [0137] determining a confidence of the latitude and longitude; and
 - [0138] determining a trustworthiness score based on the correlating, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude.
- [0139] Clause 8. An apparatus comprising:
- [0140] at least one transceiver;
 - [0141] at least one memory; and
 - [0142] at least one processor communicatively coupled to the at least one transceiver and the at least one memory and configured to:
 - [0143] obtain an assertion of a first mobile device being at a first location at a first time;
 - [0144] obtain first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;
 - [0145] obtain second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time;
 - [0146] correlate the first indications and the second indications; and
 - [0147] perform an action in response to correlation of the first indications and the second indications.
- [0148] Clause 9. The apparatus of clause 8, wherein the at least one processor is configured to perform the action in response to the correlation of the first indications and the second indications indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.
- [0149] Clause 10. The apparatus of clause 9, wherein the action comprises storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.
- [0150] Clause 11. The apparatus of clause 8, wherein:
- [0151] the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and
 - [0152] the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.
- [0153] Clause 12. The apparatus of clause 11, wherein to correlate the first indications and the second indications the at least one processor is configured to determine whether both the first indications and the second indications include a common globally-unique cell identifier.
- [0154] Clause 13. The apparatus of clause 8, wherein to correlate the first indications and the second indications the at least one processor is configured to compare a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.
- [0155] Clause 14. The apparatus of clause 8, wherein to obtain the first indications the at least one processor is configured to receive, via the at least one transceiver, the first indications from the first mobile device, and wherein the at least one processor is further configured to:
- [0156] determine a latitude and longitude of the first mobile device based on at least some of the first indications;
 - [0157] determine a confidence of the latitude and longitude; and
 - [0158] determine a trustworthiness score based on correlation of the first indications and the second indications, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude.
- [0159] Clause 15. An apparatus comprising:
- [0160] means for obtaining an assertion of a first mobile device being at a first location at a first time;
 - [0161] means for obtaining first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;
 - [0162] means for obtaining second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time;
 - [0163] means for correlating the first indications and the second indications; and
 - [0164] means for performing an action in response to correlation of the first indications and the second indications.
- [0165] Clause 16. The apparatus of clause 15, wherein the means for performing the action comprise means for performing the action in response to the correlation of the first indications and the second indications indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.
- [0166] Clause 17. The apparatus of clause 16, wherein the means for performing the action comprise means for storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.
- [0167] Clause 18. The apparatus of clause 15, wherein:
- [0168] the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and
 - [0169] the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.

- [0170] Clause 19. The apparatus of clause 18, wherein the means for correlating comprise means for determining whether both the first indications and the second indications include a common globally-unique cell identifier.
- [0171] Clause 20. The apparatus of clause 15, wherein the means for correlating comprise means for comparing a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.
- [0172] Clause 21. The apparatus of clause 15, wherein the means for obtaining the first indications comprise means for receiving the first indications from the first mobile device, the apparatus further comprising:
- [0173] means for determining a latitude and longitude of the first mobile device based on at least some of the first indications;
- [0174] means for determining a confidence of the latitude and longitude; and
- [0175] means for determining a trustworthiness score based on the correlating, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude.
- [0176] Clause 22. A non-transitory, processor-readable medium comprising processor-readable instructions to cause at least one processor of an apparatus to:
- [0177] obtain an assertion of a first mobile device being at a first location at a first time;
- [0178] obtain first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;
- [0179] obtain second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time;
- [0180] correlate the first indications and the second indications; and
- [0181] perform an action in response to the correlation of the first indications and the second indications.
- [0182] Clause 23. The non-transitory, processor-readable medium of clause 22, wherein the processor-readable instructions to cause the at least one processor to perform the action comprise processor-readable instructions to cause the at least one processor to perform the action in response to the correlation of the first indications and the second indications indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.
- [0183] Clause 24. The non-transitory, processor-readable medium of clause 23, wherein the processor-readable instructions to cause the at least one processor to perform the action comprise processor-readable instructions to cause the at least one processor to store, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.
- [0184] Clause 25. The non-transitory, processor-readable medium of clause 22, wherein:
- [0185] the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and
- [0186] the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.
- [0187] Clause 26. The non-transitory, processor-readable medium of clause 25, wherein the processor-readable instructions to cause the at least one processor to correlate comprise processor-readable instructions to cause the at least one processor to determine whether both the first indications and the second indications include a common globally-unique cell identifier.
- [0188] Clause 27. The non-transitory, processor-readable medium of clause 22, wherein the processor-readable instructions to cause the at least one processor to correlate comprise processor-readable instructions to cause the at least one processor to compare a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.
- [0189] Clause 28. The non-transitory, processor-readable medium of clause 22, wherein the processor-readable instructions to cause the at least one processor to obtain the first indications comprise processor-readable instructions to cause the at least one processor to receive the first indications from the first mobile device, the non-transitory, processor-readable medium further comprising processor-readable instructions to cause the at least one processor to:
- [0190] determine a latitude and longitude of the first mobile device based on at least some of the first indications;
- [0191] determine a confidence of the latitude and longitude; and
- [0192] determine a trustworthiness score based on the correlating, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude.
- Other Considerations
- [0193] Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software and computers, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or a combination of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.
- [0194] As used herein, the singular forms “a,” “an,” and “the” include the plural forms as well, unless the context clearly indicates otherwise. Thus, reference to a device in the singular (e.g., “a device,” “the device”), including in the claims, includes at least one, i.e., one or more, of such devices (e.g., “a processor” includes at least one processor (e.g., one processor, two processors, etc.), “the processor”

includes at least one processor, “a memory” includes at least one memory, “the memory” includes at least one memory, etc.). The phrases “at least one” and “one or more” are used interchangeably and such that “at least one” referred-to object and “one or more” referred-to objects include implementations that have one referred-to object and implementations that have multiple referred-to objects. For example, “at least one processor” and “one or more processors” each includes implementations that have one processor and implementations that have multiple processors.

[0195] The terms “comprises,” “comprising,” “includes,” and/or “including,” as used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0196] Also, as used herein, “or” as used in a list of items (possibly prefaced by “at least one of” or prefaced by “one or more of”) indicates a disjunctive list such that, for example, a list of “at least one of A, B, or C,” or a list of “one or more of A, B, or C” or a list of “A or B or C” means A, or B, or C, or AB (A and B), or AC (A and C), or BC (B and C), or ABC (i.e., A and B and C), or combinations with more than one feature (e.g., AA, AAB, ABBC, etc.). Thus, a recitation that an item, e.g., a processor, is configured to perform a function regarding at least one of A or B, or a recitation that an item is configured to perform a function A or a function B, means that the item may be configured to perform the function regarding A, or may be configured to perform the function regarding B, or may be configured to perform the function regarding A and B. For example, a phrase of “a processor configured to measure at least one of A or B” or “a processor configured to measure A or measure B” means that the processor may be configured to measure A (and may or may not be configured to measure B), or may be configured to measure B (and may or may not be configured to measure A), or may be configured to measure A and measure B (and may be configured to select which, or both, of A and B to measure). Similarly, a recitation of a means for measuring at least one of A or B includes means for measuring A (which may or may not be able to measure B), or means for measuring B (and may or may not be configured to measure A), or means for measuring A and B (which may be able to select which, or both, of A and B to measure). As another example, a recitation that an item, e.g., a processor, is configured to at least one of perform function X or perform function Y means that the item may be configured to perform the function X, or may be configured to perform the function Y, or may be configured to perform the function X and to perform the function Y. For example, a phrase of “a processor configured to at least one of measure X or measure Y” means that the processor may be configured to measure X (and may or may not be configured to measure Y), or may be configured to measure Y (and may or may not be configured to measure X), or may be configured to measure X and to measure Y (and may be configured to select which, or both, of X and Y to measure).

[0197] As used herein, unless otherwise stated, a statement that a function or operation is “based on” an item or condition means that the function or operation is based on the stated item or condition and may be based on one or more items and/or conditions in addition to the stated item or condition.

[0198] Substantial variations may be made in accordance with specific requirements. For example, customized hardware might also be used, and/or particular elements might be implemented in hardware, software (including portable software, such as applets, etc.) executed by a processor, or both. Further, connection to other computing devices such as network input/output devices may be employed. Components, functional or otherwise, shown in the figures and/or discussed herein as being connected or communicating with each other are communicatively coupled unless otherwise noted. That is, they may be directly or indirectly connected to enable communication between them.

[0199] The systems and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

[0200] A wireless signaling system is one in which at least some signals are conveyed wirelessly, i.e., by electromagnetic and/or acoustic waves propagating through atmospheric space rather than through a wire or other physical connection, between wireless devices (devices configured for at least wireless signal transfer). A wireless signaling system (also called a wireless signaling network) may not have all signals transmitted wirelessly, but is configured to have at least some signals transmitted wirelessly. Further, the term “wireless signaling device,” or similar term, does not require that signal transfer using the wireless signaling device is exclusively, or even primarily, wireless, or that the device be a mobile device, but indicates that the device includes wireless signal transfer capability (one-way or two-way), e.g., includes at least one radio (each radio being part of a transmitter, receiver, or transceiver) for wireless signal transfer.

[0201] Specific details are given in the description herein to provide a thorough understanding of example configurations (including implementations). However, configurations may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. The description herein provides example configurations, and does not limit the scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations provides a description for implementing described techniques. Various changes may be made in the function and arrangement of elements.

[0202] The terms “processor-readable medium,” “machine-readable medium,” and “computer-readable medium,” as used herein, refer to any medium that participates in providing data that causes a machine to operate in a specific fashion. Using a computing platform, various processor-readable media might be involved in providing instructions/code to processor(s) for execution and/or might be used to store and/or carry such instructions/code (e.g., as signals). In many implementations, a processor-readable medium is a physical and/or tangible storage medium. Such a medium may take many forms, including but not limited to, non-volatile media and volatile media. Non-volatile

media include, for example, optical and/or magnetic disks. Volatile media include, without limitation, dynamic memory.

[0203] Having described several example configurations, various modifications, alternative constructions, and equivalents may be used. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the disclosure. Also, a number of operations may be undertaken before, during, or after the above elements are considered. Accordingly, the above description does not bound the scope of the claims.

[0204] Unless otherwise indicated, “about” and/or “approximately” as used herein when referring to a measurable value such as an amount, a temporal duration, and the like, encompasses variations of $\pm 20\%$ or $\pm 10\%$, $\pm 5\%$, or $\pm 0.1\%$ from the specified value, as appropriate in the context of the systems, devices, circuits, methods, and other implementations described herein. Unless otherwise indicated, “substantially” as used herein when referring to a measurable value such as an amount, a temporal duration, a physical attribute (such as frequency), and the like, also encompasses variations of $\pm 20\%$ or $\pm 10\%$, $\pm 5\%$, or $\pm 0.1\%$ from the specified value, as appropriate in the context of the systems, devices, circuits, methods, and other implementations described herein.

[0205] A statement that a value exceeds (or is more than or above) a first threshold value is equivalent to a statement that the value meets or exceeds a second threshold value that is slightly greater than the first threshold value, e.g., the second threshold value being one value higher than the first threshold value in the resolution of a computing system. A statement that a value is less than (or is within or below) a first threshold value is equivalent to a statement that the value is less than or equal to a second threshold value that is slightly lower than the first threshold value, e.g., the second threshold value being one value lower than the first threshold value in the resolution of a computing system.

1. A mobile device location assertion verification method comprising:

- obtaining, at an apparatus, an assertion of a first mobile device being at a first location at a first time;
- obtaining, at the apparatus, first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;
- obtaining, at the apparatus, second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time;
- correlating, at the apparatus, the first indications and the second indications; and
- performing an action at the apparatus in response to the correlating.

2. The method of claim 1, wherein performing the action comprises performing the action in response to the correlating indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.

3. The method of claim 2, wherein performing the action comprises storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.

4. The method of claim 1, wherein:

the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and

the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.

5. The method of claim 4, wherein the correlating comprises determining whether both the first indications and the second indications include a common globally-unique cell identifier.

6. The method of claim 1, wherein the correlating comprises comparing a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.

7. The method of claim 1, wherein obtaining the first indications comprises receiving the first indications at the apparatus from the first mobile device, the method further comprising:

- determining a latitude and longitude of the first mobile device based on at least some of the first indications;
- determining a confidence of the latitude and longitude; and

determining a trustworthiness score based on the correlating, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude.

8. An apparatus comprising:

- at least one transceiver;
- at least one memory; and
- at least one processor communicatively coupled to the at least one transceiver and the at least one memory and configured to:

obtain an assertion of a first mobile device being at a first location at a first time;

obtain first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;

obtain second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time;

correlate the first indications and the second indications; and

perform an action in response to correlation of the first indications and the second indications.

9. The apparatus of claim 8, wherein the at least one processor is configured to perform the action in response to the correlation of the first indications and the second indications indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.

10. The apparatus of claim 9, wherein the action comprises storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.

11. The apparatus of claim **8**, wherein:

the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and

the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.

12. The apparatus of claim **11**, wherein to correlate the first indications and the second indications the at least one processor is configured to determine whether both the first indications and the second indications include a common globally-unique cell identifier.

13. The apparatus of claim **8**, wherein to correlate the first indications and the second indications the at least one processor is configured to compare a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.

14. The apparatus of claim **8**, wherein to obtain the first indications the at least one processor is configured to receive, via the at least one transceiver, the first indications from the first mobile device, and wherein the at least one processor is further configured to:

determine a latitude and longitude of the first mobile device based on at least some of the first indications; determine a confidence of the latitude and longitude; and determine a trustworthiness score based on correlation of the first indications and the second indications, the trustworthiness score being indicative of a trustworthiness of the latitude, the longitude, and the confidence of the latitude and longitude.

15. An apparatus comprising:

means for obtaining an assertion of a first mobile device being at a first location at a first time;

means for obtaining first indications of a radio frequency environment corresponding to the first mobile device, the first location, and the first time;

means for obtaining second indications of a second radio frequency environment corresponding to a second mobile device, a second location, and a second time; means for correlating the first indications and the second indications; and

means for performing an action in response to correlation of the first indications and the second indications.

16. The apparatus of claim **15**, wherein the means for performing the action comprise means for performing the action in response to the correlation of the first indications and the second indications indicating that the first mobile device is greater than fifty percent likely to have been within a threshold distance of the first location within a threshold time of the first time.

17. The apparatus of claim **16**, wherein the means for performing the action comprise means for storing, in a blockchain ledger, a location verification indication indicating that the first mobile device was at least within the threshold distance of the first location at least within the threshold time of the first time.

18. The apparatus of claim **15**, wherein:

the first indications include a first serving cell identifier of a first serving cell of the first mobile device at the first time, and a first neighbor cell identifier of a first neighbor cell corresponding to the first mobile device at the first time; and

the second indications include a second serving cell identifier of a second serving cell of the second mobile device at the second time, and a second neighbor cell identifier of a second neighbor cell corresponding to the second mobile device at the second time.

19. The apparatus of claim **18**, wherein the means for correlating comprise means for determining whether both the first indications and the second indications include a common globally-unique cell identifier.

20. The apparatus of claim **15**, wherein the means for correlating comprise means for comparing a first signal-to-noise ratio of the first indications to a second signal-to-noise ratio of the second indications, the first signal-to-noise ratio corresponding to a first measurement of a first signal from a signal source, and the second signal-to-noise ratio corresponding to a second measurement of a second signal from the signal source.

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